5G RAN

5G Networking and Signaling Feature Parameter Description

Issue Draft A

Date 2025-12-31

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Contents

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 03 (2025-11-07).

- Technical changes

  None

- Editorial changes

  Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-021104	SA Option 2 Architecture	4 Basic Signaling Procedures in SA Networking

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Basic signaling procedures in SA networking	Supported only in SA networking	4 Basic Signaling Procedures in SA Networking
Basic signaling procedures in NSA networking	Supported only in NSA networking	5 Basic Signaling Procedures in NSA Networking

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Basic signaling procedures in SA networking	Supported in both high and low frequency bands, with the following differences: • OSI payload reduction is supported only in high frequency bands. • SA networking is supported only in FWA scenarios in high frequency bands and is supported in all scenarios in low frequency bands. • Allocation of dedicated physical random access channel (PRACH) resources by gNodeB for other system information (OSI) is supported only in low frequency bands. • RRC_INACTIVE state is supported only in low frequency bands.	**4 Basic Signaling Procedures in SA Networking**
Basic signaling procedures in NSA networking	Optimized counter measurement for contention-based RA: Supported only in low frequency bands Other functions: None	**5 Basic Signaling Procedures in NSA Networking**

3 Overview

5G networking modes include standalone (SA) and non-standalone (NSA), as defined in section 7.2 "5G Architecture Options" of 3GPP TR 38.801 V14.0.0. Huawei supports the following networking modes:

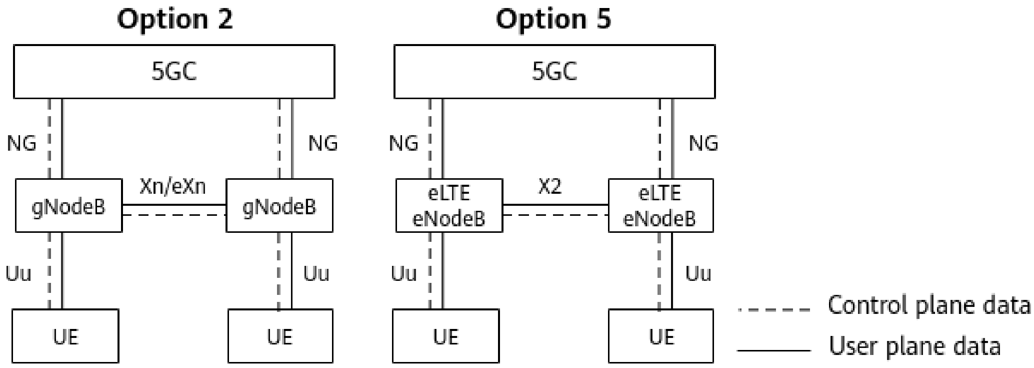
- SA networking: For its description, see **SA Networking**. For the relevant signaling procedures, see **4 Basic Signaling Procedures in SA Networking**.
- NSA networking: For its description, see **NSA Networking**. For the relevant signaling procedures, see **5 Basic Signaling Procedures in NSA Networking**.
- NSA and SA hybrid networking: For its description, see **NSA and SA Hybrid Networking**. For the relevant signaling procedures related to:
  - Access of 5G-capable UEs to SA networks, see **4 Basic Signaling Procedures in SA Networking**.
  - Access of EN-DC-capable UEs to NSA networks, see **5 Basic Signaling Procedures in NSA Networking**.

SA Networking

In SA networking, gNodeBs or enhanced 4G base stations (referred to as eLTE eNodeBs) are connected to the 5G Core Network (5GC) in a standalone manner. It is the target networking of 5G network development. For details about eLTE eNodeBs, see section 3.1 "Definitions" in 3GPP TR 38.801 V14.0.0.

Figure 3-1 shows the SA architecture options.

Figure 3-1 SA architecture options



Huawei SA networking uses the Option 2 architecture. Option 2 is an end-to-end 5G network architecture, in which the UEs, new radio, and core network all comply with 5G standards. Option 2 includes the following elements:

- 5GC: includes the access and mobility management function (AMF) and the user plane function (UPF). AMF provides UE access permission and mobility management, while UPF provides user plane management.
- gNodeB: consists of a baseband unit, radio equipment, and antennas. It is used for service data and signaling transmission.
- UE: is a 5G terminal.

The 5GC is connected to the gNodeBs through the NG interface. The gNodeBs are connected to each other through the Xn interface, and to the UEs through the Uu interface. **Table 3-1** describes these interfaces.

Table 3-1 Interfaces in SA networking

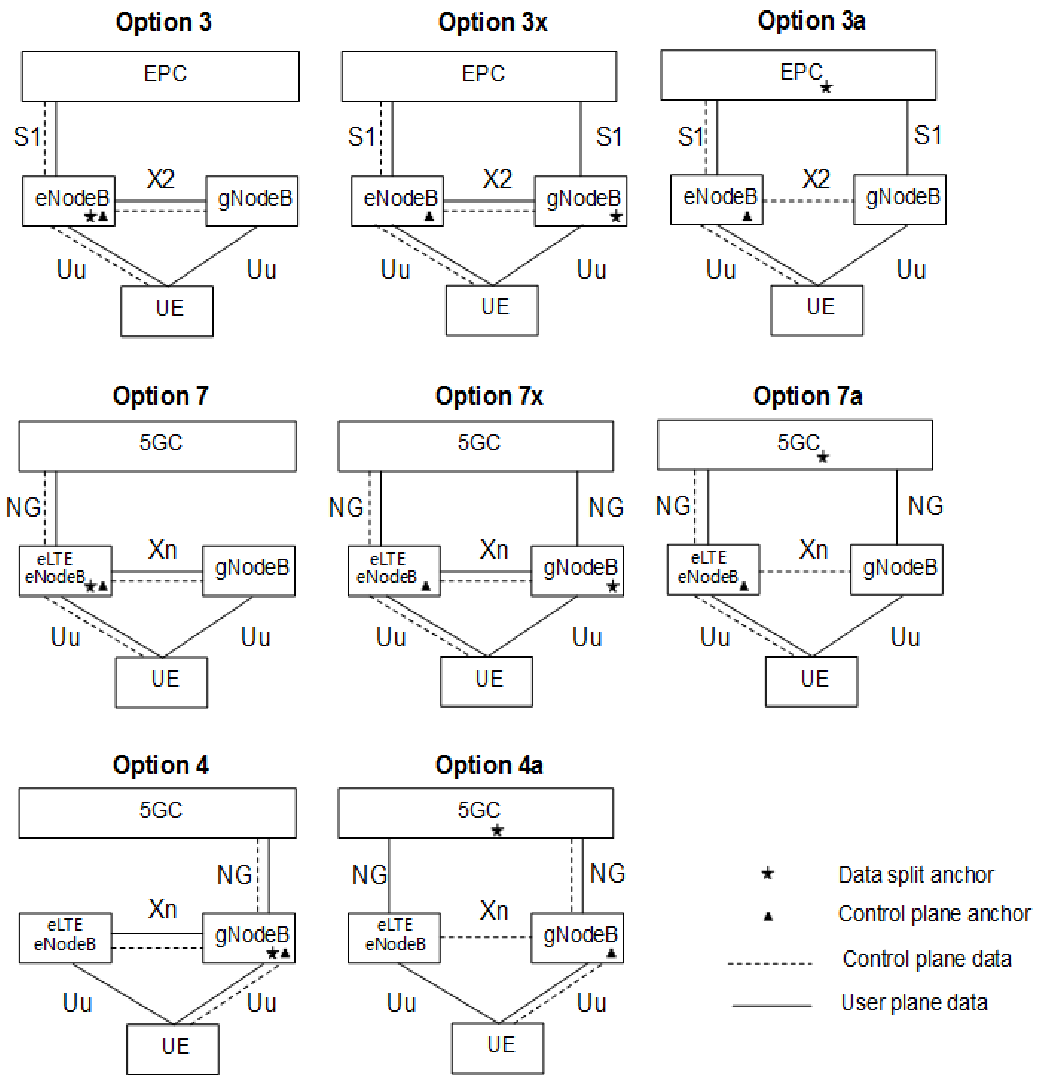
Interface	Description	Reference
NG	Consists of NG-C and NG-U interfaces, which implement NG control plane functions and NG user plane functions, respectively.	*NG and Xn Self-Management*
Xn	Consists of Xn-C and Xn-U interfaces, which implement Xn control plane functions and Xn user plane functions, respectively.	*NG and Xn Self-Management*
eXn	Consists of eXn-C and eXn-U interfaces, which implement eXn control plane functions and eXn user plane functions, respectively.	*eXn Self-Management*
Uu	Consists of Uu-C and Uu-U interfaces, which implement Uu control plane functions and Uu user plane functions, respectively.	N/A

NSA Networking

In NSA networking, 5G networks are built on and integrated with 4G networks. Signals are transmitted using radio resources from both eNodeBs and gNodeBs. Such an approach can help fast commercial use of 5G networks.

Figure 3-2 shows the NSA architecture options.

Figure 3-2 NSA architecture options



Huawei NSA networking uses the Option 3 and Option 3x architectures. In both architectures, the eNodeB serves as the control plane anchor to carry control plane data. Their difference is as follows:

- In Option 3, the eNodeB is the data split anchor. The eNodeB distributes some of the user plane data to the gNodeB, and still carries the remaining user plane data.
- In Option 3x, the gNodeB is the data split anchor. The gNodeB distributes some of the user plane data to the eNodeB, and still carries the remaining user plane data.

Option 3 and Option 3x include the following elements:

- EPC: is the 4G core network, which provides functions such as mobility management and user plane management.
- eNodeB: serves as the master eNodeB (MeNB) in Option 3 and Option 3x.
- gNodeB: serves as the secondary gNodeB (SgNB) in Option 3 and Option 3x.

- UE: is a terminal capable of E-UTRA-NR dual connectivity (EN-DC). The UE maintains DC with the eNodeB and the gNodeB, and uses radio resources from both base stations for transmission.

The EPC is connected to the eNodeB and gNodeB through the S1 interface. The eNodeB and the gNodeB are connected to each other through the X2 interface. The gNodeB is connected to the UE through the Uu interface. **Table 3-2** describes these interfaces.

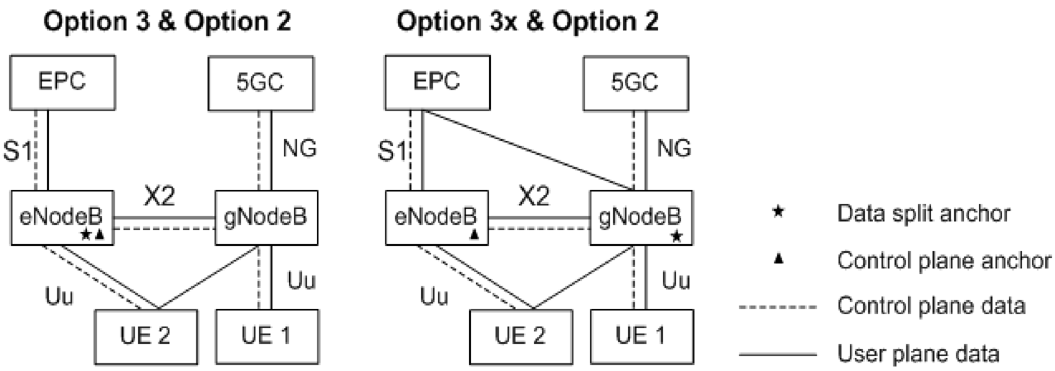
Table 3-2 Interfaces in NSA networking

Interface	Description	Reference
S1	Consists of S1-C and S1-U interfaces, which implement S1 control plane functions and S1 user plane functions, respectively.	*S1 and X2 Self-Management* in eRAN feature documentation
X2	Consists of X2-C and X2-U interfaces, which implement X2 control plane functions and X2 user plane functions, respectively.	*S1 and X2 Self-Management* in eRAN feature documentation
Uu	Consists of Uu-C and Uu-U interfaces, which implement Uu control plane functions and Uu user plane functions, respectively.	N/A

NSA and SA Hybrid Networking

Huawei supports NSA and SA hybrid networking, which facilitates smooth evolution from NSA to SA. **Figure 3-3** shows the NSA and SA hybrid networking architecture options.

Figure 3-3 NSA and SA hybrid networking architecture options



Logically, there are two networks in NSA and SA hybrid networking. For details about Option 3 and Option 3x, see **NSA Networking**. For details about Option 2, see **SA Networking**.

4 Basic Signaling Procedures in SA Networking

Figure 4-1 shows the basic signaling procedures in SA networking.

Figure 4-1 Basic signaling procedures in SA networking

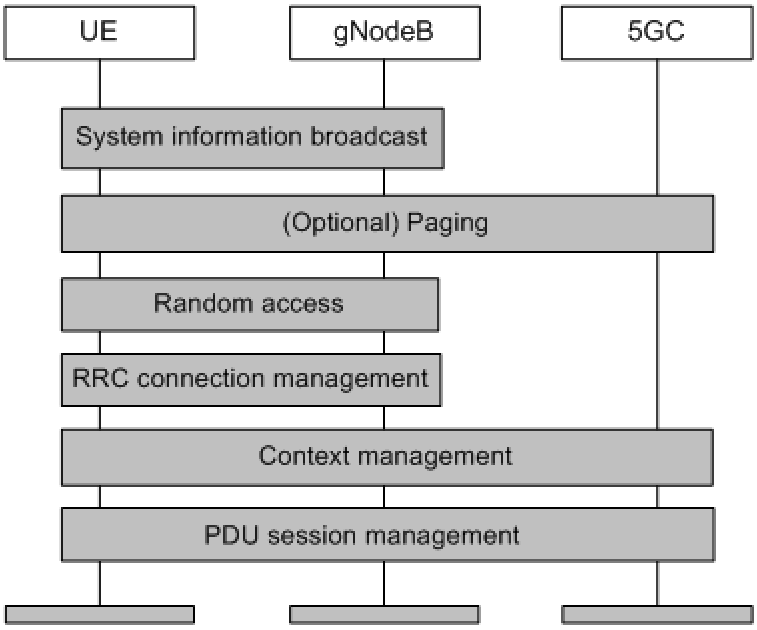


Table 4-1 describes the basic signaling procedures in SA networking.

Table 4-1 Basic signaling procedures in SA networking

Procedure	Description	Reference
System information broadcast	It is the first step for a UE to obtain the basic network service information. Through this procedure, the UE can obtain the basic access stratum (AS) and non-access stratum (NAS) information.	**4.1.1 System Information Broadcast**
(Optional) Paging	When the network needs to set up a connection with a UE, the network initiates a paging procedure to find the UE. This procedure involves the terminating UE, but not the originating UE.	**4.1.2 Paging**
Random access (RA)	RA starts when a UE sends an RA preamble and ends when an RRC connection is set up between the UE and the network.	**4.1.3 Random Access**
RRC connection management	RRC connection management includes RRC connection setup, reconfiguration, release, and reestablishment between a UE and a gNodeB, as well as uplink out-of-synchronization management and UE inactivity management.	**4.1.4 RRC Connection Management**
Context management	The gNodeB sends the 5GC an INITIAL UE MESSAGE to trigger NG-C connection setup and UE context reception. Context management includes UE context setup, modification, and release.	**4.1.5 Context Management**
PDU session management	A PDU session is a data connection between a UE and a data network (DN). PDU session management includes PDU session setup, modification, and release.	**4.1.6 PDU Session Management**

4.1 Principles

4.1.1 System Information Broadcast

System information broadcast is the first step for a UE to obtain the basic network service information. In this procedure, the gNodeB transmits system information and the UE obtains system information.

- The content of system information broadcast is carried in system information blocks (SIBs). For details about SIBs, see **4.1.1.1 System Information Block**.
- For details about how the gNodeB transmits system information, see **4.1.1.2 System Information Transmission**.
- For details about how a UE receives system information, see **4.1.1.3 System Information Acquisition**.

4.1.1.1 System Information Block

According to section 7.3 "System Information Handling" of 3GPP TS 38.300 V15.5.0, system information can be classified into minimum system information (MSI) and other system information (OSI) by content.

- **MSI includes the master information block (MIB) and System Information Block 1 (SIB1). The MIB provides the information used to acquire SIB1, and SIB1 provides basic information required for cell selection when a UE initially accesses the network.**
- **OSI includes SIB2 to SIB*n*. It provides information such as the mobility, time, earthquake and tsunami warning system (ETWS), and commercial mobile alert system (CMAS) for a UE.** Only SIB2, SIB3, SIB4, SIB5, and SIB8 are currently supported.

The MIB and SIBs contain different information, as listed in **Table 4-2**. For details, see sections 6.2.2 "Message definitions" and 6.3.1 "System information blocks" of 3GPP TS 38.331 V15.5.0.

Table 4-2 Contents of the MIB and SIBs

Category	Information Type	Content
MSI	MIB	System frame number (SFN) and information used to acquire SIB1
	SIB1	Operator information, initial bandwidth part (BWP) information[a], and scheduling information of other SIBs of the cell
OSI	SIB2	Common information required for intra-frequency cell reselection, inter-frequency cell reselection, and inter-RAT cell reselection
	SIB3	Intra-frequency cell reselection parameters and intra-frequency cell blacklist
	SIB4	Inter-frequency cell reselection parameters (including information about non-serving intra-RAT frequencies and inter-frequency neighboring cells) and inter-frequency cell blacklist
	SIB5	Inter-RAT cell reselection parameters (including information about inter-RAT frequencies and inter-RAT neighboring cells) and inter-RAT cell blacklist
	SIB8	CMAS notification

Category	Information Type	Content
a: In SA networking, the initial BWP information is carried in the *locationAndBandwidth* field of the *initialDownlinkBWP* and *initialUplinkBWP* IEs in SIB1. Full-bandwidth initial BWP configuration is controlled by the **INIT_BWP_FULL_BW_SW** option (selected by default) of the **NRDUCellAlgoSwitch**.***BwpConfigPolicySwitch*** parameter. Changing the setting of this option will cause the cell to restart, affecting admitted UEs. ● If this option is selected, the gNodeB configures the full-bandwidth initial BWP for all UEs. ● If this option is deselected, the gNodeB configures an initial BWP with the bandwidth specified by CORESET#0 for all UEs. UEs cannot use the full bandwidth during access, affecting the UE access success rate. In addition, PUCCH and PRACH resources in the initial BWP interrupt the continuity of PUSCH resources in the frequency domain, affecting the uplink cell throughput. For details about CORESET#0, see section 6.3.2 "Radio resource control information elements" of 3GPP TS 38.331 V15.5.0.		

4.1.1.2 System Information Transmission

Different types of system information are transmitted over different channels (as shown in **Figure 4-2**):

MIB: BCCH (broadcast control channel, a logical channel) -> BCH (broadcast channel, a transport channel) -> PBCH (physical broadcast channel, a physical channel)

- SIB1: BCCH (a logical channel) -> DL-SCH (downlink shared channel, a transport channel) -> PDSCH (physical downlink shared channel, a physical channel)
- OSI: BCCH (a logical channel) -> DL-SCH (a transport channel) -> PDSCH (a physical channel) (Among OSI, multiple SIBs with the same scheduling period are encapsulated into one SI message for transmission.)

Figure 4-2 System information transmission paths

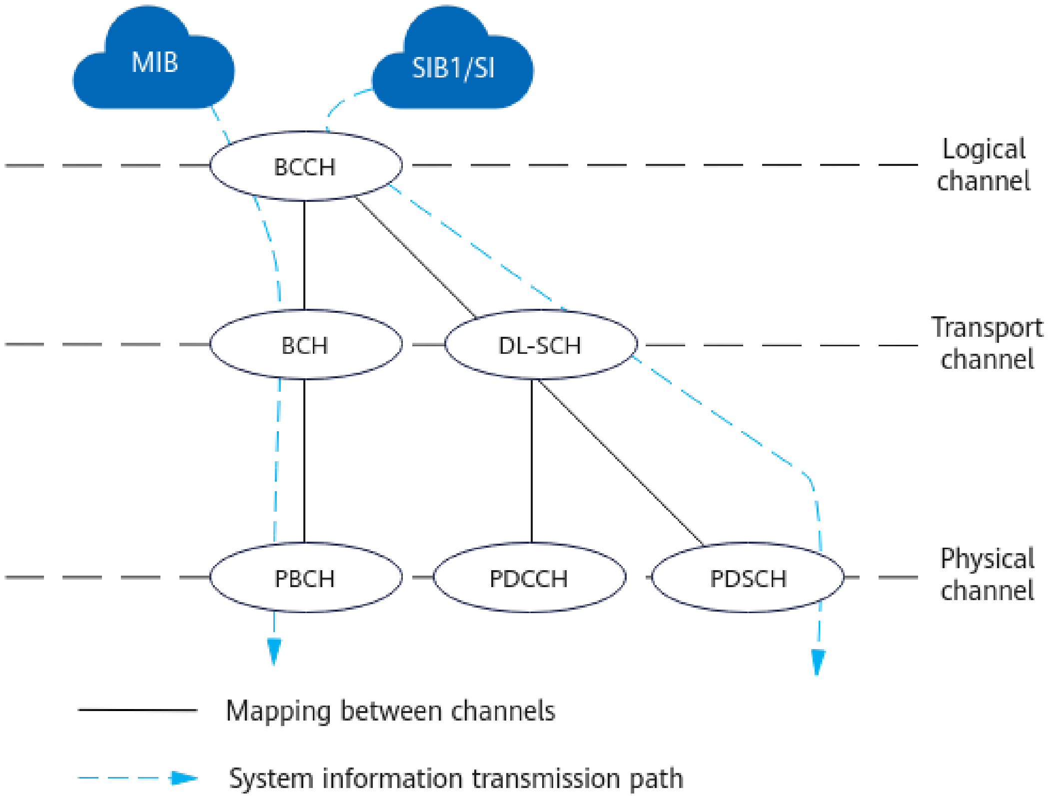


Table 4-3 describes how system information is transmitted.

Table 4-3 System information transmission

System Information	Transmission	Parameter
MIB	The gNodeB periodically and repeatedly broadcasts it within a scheduling period (80 ms).	The broadcast period is specified by the **NRDUCell.*SsbPeriod*** parameter[a].
SIB1	The gNodeB periodically and repeatedly broadcasts it within a scheduling period (160 ms).	The broadcast period is specified by the **NRDUCell.*Sib1Period*** parameter.
OSI	The gNodeB broadcasts it in a scheduling period (indicated by SIB1) either periodically and repeatedly or on demand, based on the system information type, broadcast mode, and broadcast period.	The system information type is specified by the **gNBSibConfig**.***SibType*** parameter. The broadcast mode is specified by the **gNBSibConfig**.***SibTransPolicy*** parameter.

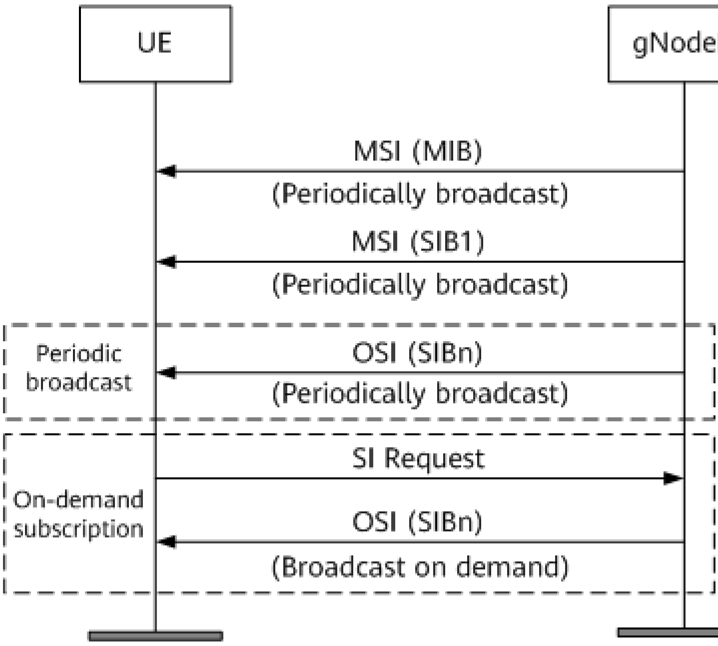
System Information	Transmission	Parameter
		The broadcast period is specified by the **gNBSibConfig**.***SibTransPeriod*** parameter.
a: If this parameter is set to **MS160**, the broadcast period is longer than the scheduling period, so that broadcast messages cannot be sent to all UEs within a scheduling period. In this case, the MIB is transmitted at an interval of 160 ms.		

NOTE

The **NRDUCell**.***SibConfigId*** and **gNBSibConfig**.***SibConfigId*** parameters can be used to specify the mode and period of system information broadcast in cells.

Figure 4-3 shows system information broadcast.

Figure 4-3 System information broadcast



1. A Short Message indicating a system information change is sent to the UE.

   The gNodeB sets the *systemInfoModification* field in the Short Message to **1**. For details, see section 6.5 "Short Message" in 3GPP TS 38.331 V15.5.0.

2. The updated system information is sent in the next BCCH modification period.

   The gNodeB changes the *valueTag* value of the corresponding system information in SIB1, and sends the updated system information to the UE in the next BCCH modification period.

System Information Update

As described in section 6.3.2 "Radio resource control information elements" of 3GPP TS 38.331 V15.5.0, the system information update process is performed within specific radio frames, which are referred to as the BCCH modification period (*m* radio frames).

The BCCH modification period is equal to *A* x *B*, where:

- *A* denotes *modificationPeriodCoeff*, which means the coefficient for the modification period. It indicates the minimum number of times the UE monitors paging messages within the BCCH modification period. It is always set to **2** and is not configurable.
- *B* denotes *defaultPagingCycle*, which means the default paging cycle, expressed in units of radio frames. It is specified by the **NRDUCellPagingConfig**.***DefaultPagingCycle*** parameter.

They are both included in SIB1. The BCCH modification period boundaries are defined by SFN values for which the result of SFN mod *m* is 0. That is, the SFN of the start frame for a period must meet the requirement of "SFN mod *m* = 0".

OSI Payload Reduction

When power aggregation takes effect in a high-frequency TDD cell, the OSI payload reduction function can be used to improve OSI demodulation performance on the UE side. This function is controlled by the **OSI_PAYLOAD_REDUCE_SW** option of the **NRDUCellDlSch**.***DlSchAlgoSwitch*** parameter. With this option selected, OSI conveys fewer bits, so that fewer RBs are used for each OSI transmission.

4.1.1.3 System Information Acquisition

A UE obtains system information in the following order:

1. MIB: The BCH transport format is predefined. Therefore, the UE receives the MIB without obtaining any other information from the network.
2. SIB1: After receiving the MIB, the UE receives SIB1 according to the configuration specified by the MIB.
3. OSI: After receiving SIB1, the UE receives OSI according to the configuration specified by SIB1.
   - If SIB1 indicates that the OSI is delivered in broadcast mode, the UE receives this OSI in the corresponding SI window.
   - If SIB1 indicates that the OSI is delivered in on-demand delivery mode, the UE receives this OSI in a way depending on whether SI-RequestConfig is configured in SIB1.
     - If SI-RequestConfig is configured, the UE requests the OSI through Msg1. In this mode, the gNodeB allocates dedicated physical random access channel (PRACH) resources for the OSI. This mode applies when PRACH resources are sufficient.
     - If SI-RequestConfig is not configured, the UE requests the OSI through Msg3. In this mode, the gNodeB does not allocate dedicated PRACH resources for the OSI. This mode applies when PRACH resources are insufficient.

NOTE

To prevent a UE from repeatedly sending requests, if SI-RequestConfig is configured, the gNodeB acknowledges the reception of a request through Msg2 and immediately broadcasts the requested OSI. If SI-RequestConfig is not configured, the gNodeB acknowledges the reception of a request through Msg4 and immediately broadcasts the requested OSI.

When a UE enters a new serving cell, the UE obtains cell system information in the following scenarios:

- The UE selects the cell after being powered on.
- The UE is to reselect to the new serving cell.
- The UE completes a handover process.
- The UE is transferred to the NG-RAN from another RAT.
- The UE returns from a non-coverage area to a coverage area.

To save power, the UE does not repeatedly obtain system information during each system information broadcast period. Instead, the UE re-obtains serving cell system information only when the serving cell broadcast parameters change or the system information validity period expires. The UE re-obtains the cell system information in the following scenarios:

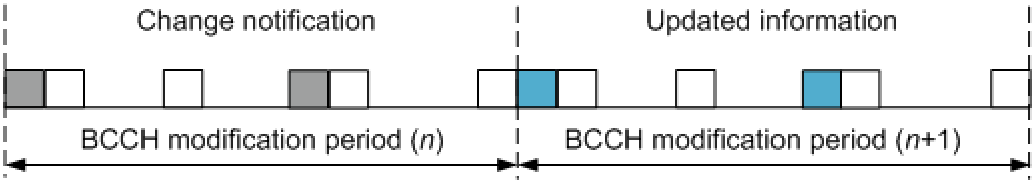
- The UE receives a system information change notification in a Short Message from the gNodeB.

  The UE reads the valueTag value of system information in SIB1 and compares it with the one acquired last time. If the value has changed, the UE learns that the system information has changed, and then re-obtains the system information. If the value has not changed, the UE learns that the system information has not changed, and will not re-obtain the system information.
- The system information validity period expires.

  The system information stored on the UE is only valid for 3 hours. After 3 hours, the UE reads all of the system information regardless of whether the valueTag value changes.

When the system information changes, the UE performs the following operations, as shown in **Figure 4-4**.

Figure 4-4 System information update procedure



1. The UE receives the Short Message in the current BCCH modification period (n).
2. The UE receives updated system information in the next BCCH modification period (n+1).

In the figure, gray and blue blocks indicate the same system information type. A color change indicates content change. The white system information blocks remain unchanged during the procedure.

4.1.2 Paging

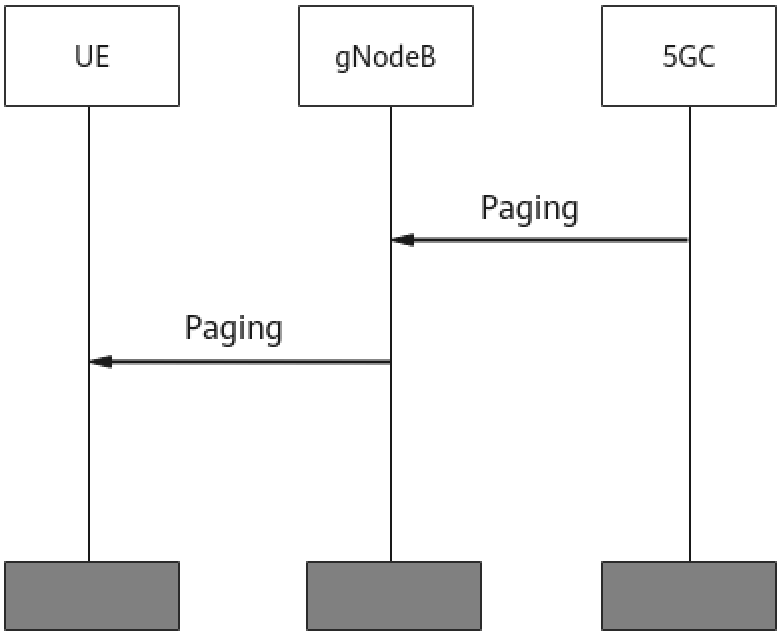
4.1.2.1 Triggering of Paging

The network searches for UEs by paging. There are two paging types, which differ in triggering sources.

- 5GC paging: triggered by the 5GC. When downlink data needs to be sent to a UE in the RRC_IDLE state, the 5GC triggers paging for the UE.
- RAN paging: triggered by the gNodeB. When downlink data needs to be sent to a UE in the RRC_INACTIVE state, the gNodeB triggers paging for the UE.

Figure 4-5 shows the message transmission for a paging procedure triggered by the 5GC.

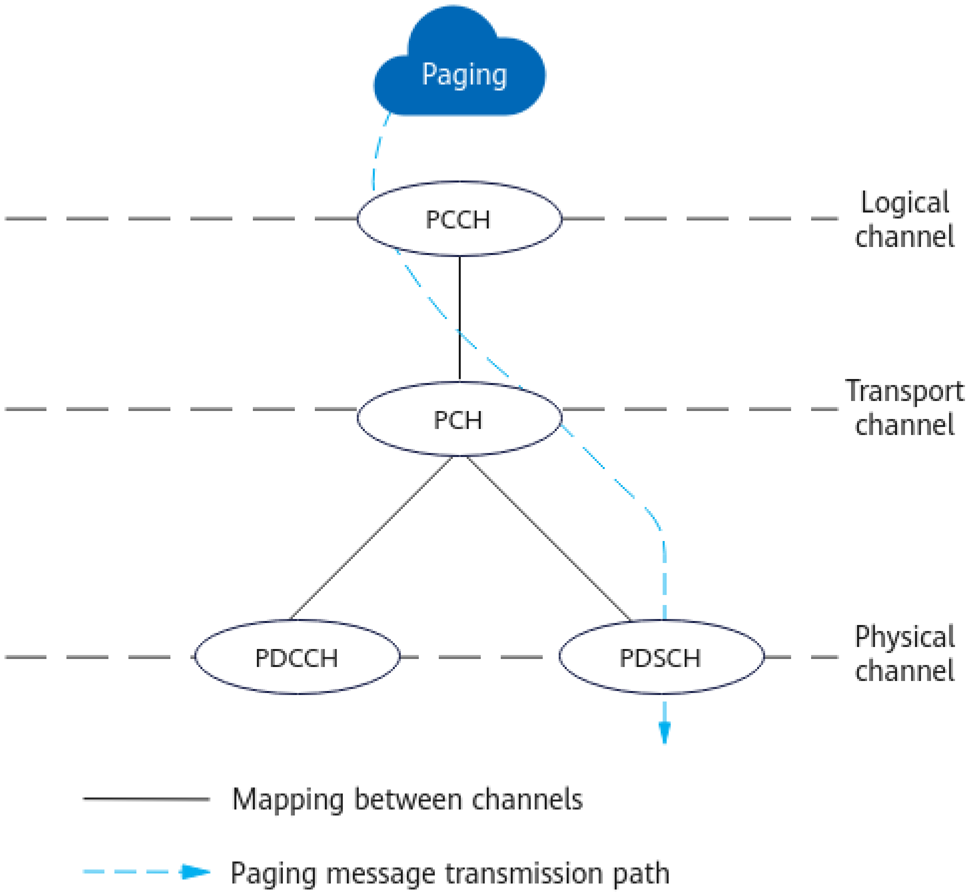
Figure 4-5 Paging message transmission



4.1.2.2 Paging Mechanism over the Air Interface

Figure 4-6 shows the paging message transmission path: PCCH (paging control channel, a logical channel) -> PCH (paging channel, a transport channel) -> PDSCH (a physical channel).

Figure 4-6 Paging message transmission path



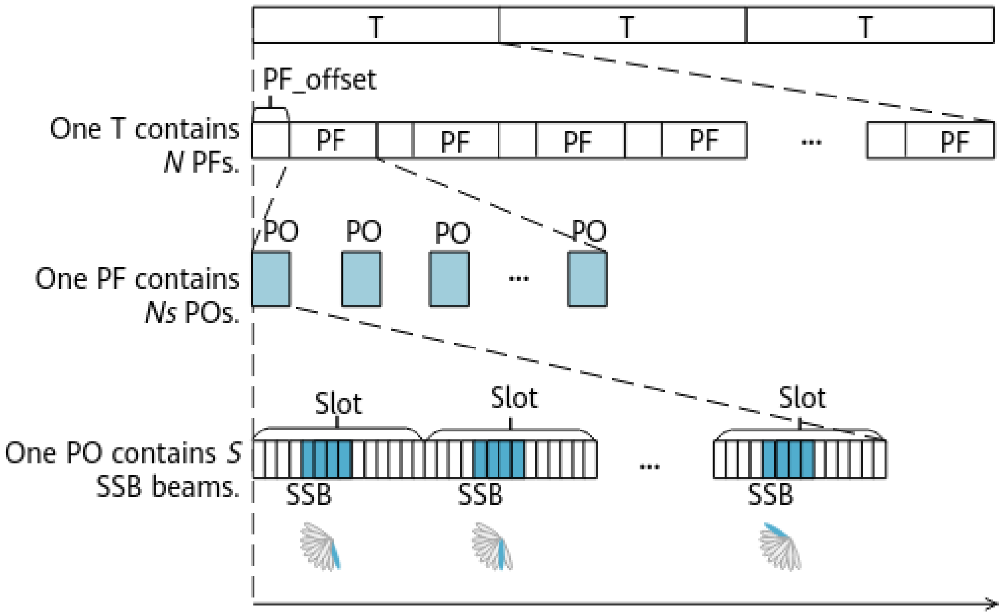
The gNodeB can schedule UEs based on the first in first out (FIFO) policy and priority-based policy. (For details about paging priorities, see section 5.4 "3GPP access specific aspects" in 3GPP TS 23.501 V0.5.0.) The policy to use for paging transmission over the air interface is determined by the **NRDUCellPagingConfig**.***PagingTransPolicy*** parameter. **Table 4-4** describes the policies.

Table 4-4 Paging transmission policies

Policy	Parameter Setting	Principles	Comparison
FIFO	The **NRDUCellPagingConfig**.***PagingTransPolicy*** parameter is set to **FIFO**.	The paging messages that reach the gNodeB earlier on a paging occasion are preferentially combined into one paging message. (A maximum of 32 paging messages can be combined.) The gNodeB then delivers the combined paging message to UEs on this paging occasion.	• Advantages: Lower-priority paging messages are not discarded. Instead, the transmission strictly follows the FIFO rule. • Disadvantages: When there are a large number of paging messages, higher-priority paging messages may not be promptly transmitted, and therefore their delay may increase.
Priority-based	The **NRDUCellPagingConfig**.***PagingTransPolicy*** parameter is set to **PRI_DIFFERENTIATED**.	The paging messages with higher priorities on a paging occasion are preferentially combined into one paging message. (A maximum of 32 paging messages can be combined.) The gNodeB then delivers the combined paging message to UEs on this paging occasion.	• Advantages: The wait time before transmitting higher-priority paging messages over the air interface decreases, accelerating responses to these messages. • Disadvantages: When there are a large number of paging messages, lower-priority paging messages may be discarded.

As described in section 7.1 "Discontinuous Reception for paging" in 3GPP TS 38.304 V15.4.0, UEs in the RRC_IDLE or RRC_INACTIVE state can use discontinuous reception (DRX) to reduce power consumption. A UE receives a paging message over the air interface only at a fixed time-domain position, as shown in **Figure 4-7**.

Figure 4-7 Paging mechanism



- T: indicates the paging cycle.
- PF: indicates the paging frame, which consists of multiple POs.
- PO: indicates the paging occasion. It is a set of PDCCH monitoring opportunities and consists of multiple slots. One PO involves *S* synchronization signal and PBCH block (SSB) beams. *S* is determined by the *ssb-PositionsInBurst* IE in SIB1. According to protocols, a low-frequency cell transmits a maximum of eight SSB beams within each PO. The number of SSB beams to be transmitted depends on factors such as slot configurations. As stipulated in section 6.2.2 "Message definitions" of 3GPP TS 38.331 V15.5.0, a maximum of 32 UEs can be paged within a PO.
- SSB: The paging messages sent on each SSB beam are the same.

The PF and PO are determined by the following formulas:

- (SFN + PF_offset) mod T = (T div N) x (UE_ID mod N)

  All radio frames whose SFN values meet the preceding formula are PFs.
- i_s of a PO = floor (UE_ID/N) mod Ns

  i_s is the index of the PO in a PF where a UE starts to receive paging messages. The *i_s*th PO is the start of a set of PDCCH monitoring occasions. With the PF and PO identified, the time when the UE receives paging messages is determined.

Huawei uses the following values of the preceding elements:

- T: indicates the paging cycle and is equal to min(Default paging cycle, UE-specific DRX cycle), where:
  – The default paging cycle is specified by the **NRDUCellPagingConfig.*DefaultPagingCycle*** parameter.
  – The UE-specific DRX cycle is carried in the *Paging DRX* IE of the paging message from the AMF to the gNodeB.

- PF_offset: indicates the frame offset of the PF and is fixed at 1.
- N: indicates the number of PFs in a paging cycle *T* and is fixed at *T*/2. The number of PFs in a paging cycle *T* is subject to the **NRDUCellPagingConfig**.***PagingFrameTransPolicy*** parameter:
  – When this parameter is set to **0**, the proportion of PFs in a paging cycle is not adjusted by this parameter.
  – When this parameter is set to **1**, the proportion of PFs in a paging cycle is 1/2.
  – When this parameter is set to **2**, the proportion of PFs in a paging cycle is 1/4.
  – When this parameter is set to **3**, the proportion of PFs in a paging cycle is 1/8.
  – When this parameter is set to **4**, the proportion of PFs in a paging cycle is 1/16.
- UE_ID: indicates the UE identity delivered by the core network and is equal to the result of 5G-S-TMSI mod 1024.
- Ns: indicates the number of POs in a PF and is fixed at 1.

4.1.2.3 Maximum Number of Paging Messages

The maximum allowed number of paging messages within a paging occasion is specified by the **NRDUCellPagingConfig**.***MaxPagingMsgNum***. This number can be dynamically adjusted. When the **NRDUCellPagingConfig**.***MaxPagingMsgNum*** parameter takes effect, the number of paging messages that can be sent within a paging occasion is equal to the smaller one between the value of this parameter and the actual number of paging messages upon requests.

The maximum number of paging messages may not take effect immediately, as affected by dynamic adjustment on the number of paging frames, which takes effect a period of time after it is enabled (using the **PAGING_FRAME_NUM_DYN_ADJ_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter). If dynamic adjustment on the number of paging frames is enabled but has not taken effect, the setting of the **NRDUCellPagingConfig**.***MaxPagingMsgNum*** parameter will not take effect until dynamic adjustment on the number of paging frames takes effect.

4.1.2.4 Paging Optimization

Paging Scheduling

Paging scheduling, enabled by selecting the **PAGING_CR_ADAP_SW** option of the **NRDUCellPagingConfig.*****PagingSchAlgoSwitch*** parameter, allows the base station to use a lower code rate for transmitting high-priority paging messages, in order to increase the transmission success rate.

- If there are high-priority paging messages within a paging occasion, paging messages are transmitted using 1/4 of the original code rate (the code rate used before this option is selected) as the initial target code rate during paging message scheduling.
- If there are no high-priority paging messages within a paging occasion, paging messages are transmitted using the code rate calculated as follows as

the initial target code rate during paging message scheduling: Original code rate x Scaling factor specified by the **NRDUCellDlSch**.***PagingAndRarTbScalFactor*** parameter.

Delayed Paging Scheduling

If dynamic adjustment on the number of paging frames is enabled (using the **PAGING_FRAME_NUM_DYN_ADJ_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter) but has not taken effect, paging scheduling (enabled using the **PAGING_CR_ADAP_SW** option of the **NRDUCellPagingConfig**.***PagingSchAlgoSwitch*** parameter) will not take effect until dynamic adjustment on the number of paging frames takes effect.

Paging Message Discard Timer

During paging scheduling, the base station checks whether each paging message has been present for a period longer than the length of the discard timer specified by the **NRDUCellPagingConfig**.***PagingTimeoutDiscardTimer*** parameter.

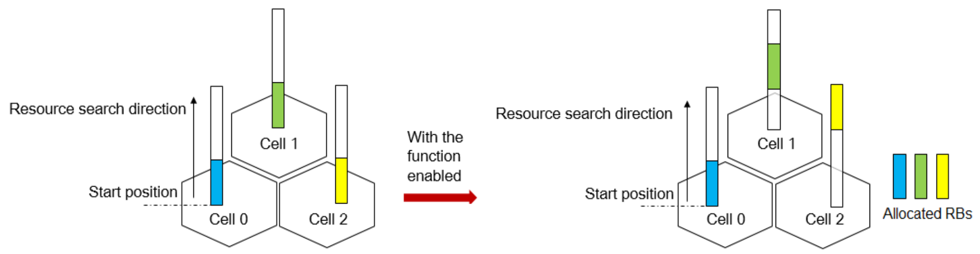
- If a paging message has, the base station discards the message.
- If a paging message has not, the base station continues to schedule the message.

Paging Interference Randomization

If paging messages are sent in a cell over only 1/3 of CORESET 0 frequency-domain resources of the cell, paging interference randomization can be used. It staggers the resources allocated in adjacent cells whenever possible, reducing interference from neighboring cells to UEs in the local cell. This function is enabled by selecting the **PAGING_INTRF_RANDOM_SW** option of the **NRDUCellPagingConfig**.***PagingSchAlgoSwitch*** parameter.

Figure 4-8 illustrates how paging interference randomization works.

Figure 4-8 Paging interference randomization



With this function, the start positions for RB allocation are staggered between adjacent cells based on the physical cell identifiers (PCIs) of the cells. RBs are allocated from different start positions in the band, reducing interference from neighboring cells to UEs in the local cell.

Paging Capacity Mode

The paging capacity mode is specified by the **NRDUCellPagingConfig**.***PagingCapacityMode*** parameter. If this parameter is set to **ENHANCE_MODE_1**, enhanced mode 1 is used, which increases the paging capacity of the cell. Otherwise, the default mode is used.

Paging Origin

The paging origin function is controlled by the **PAGING_ORIGIN_SW** option of the **gNodeBAlgo**.***PagingAlgoSwitch*** parameter. This function is enabled only if this option is selected. With this function, the base station can transfer the paging origin IE included in paging messages from the core network to UEs over the Uu interface. For details about this function, see section 6.2.2 "Message definitions" in 3GPP TS 38.331 V18.3.0.

4.1.3 Random Access

When cell search is complete, a UE achieves downlink synchronization with the cell and can receive downlink data. However, the UE has not achieved uplink synchronization with the cell yet. To achieve uplink synchronization for uplink transmission, the UE performs random access (RA) to establish a connection with the cell. During RA, the UE sends an RA preamble to initiate access on specific PRACH time-frequency resources. The RA preamble is used to inform the gNodeB of an RA request, enabling the gNodeB to estimate the transmission delay between the gNodeB and the UE. For details about RA to gNodeBs in SA networking, see *Random Access Control*.

4.1.4 RRC Connection Management

RRC connection management includes the functions described in **Table 4-5**.

Table 4-5 RRC connection management

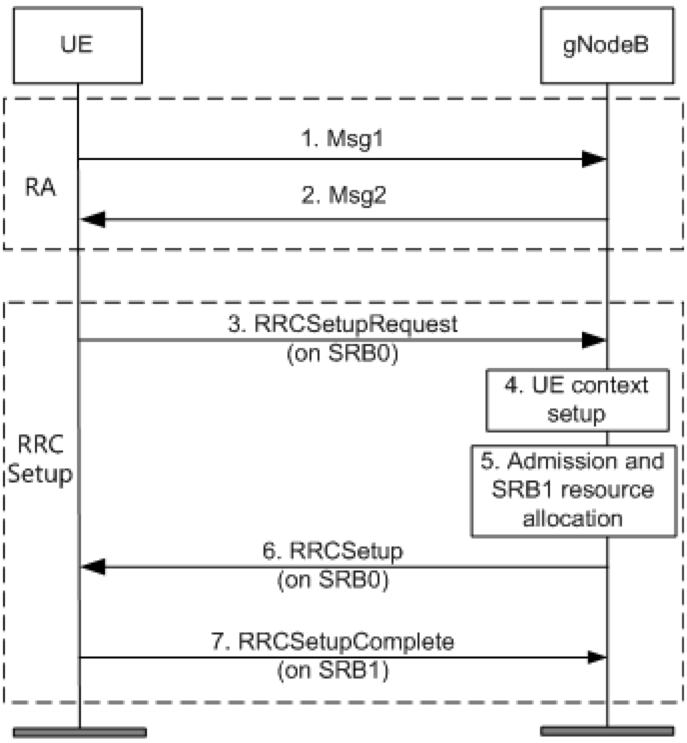
Function	Description	Reference
RRC connection setup	RRC connection setup is a procedure for establishing SRB1. After the SRB1 is established, a UE enters connected mode.	**4.1.4.1 RRC Connection Setup**
RRC connection reconfiguration	RRC connection reconfiguration is a procedure for reconfiguring signaling radio bearers (SRBs) and data radio bearers (DRBs) after RRC connection setup.	**4.1.4.2 RRC Connection Reconfiguration**
RRC connection release	RRC connection release is a procedure for releasing the RRC connection between a UE and a gNodeB, including releasing all radio bearer resources.	**4.1.4.3 RRC Connection Release**
RRC connection reestablishment	RRC connection reestablishment is a service processing procedure initiated by a UE to quickly reestablish an RRC connection.	**4.1.4.4 RRC Connection Reestablishment**

Function	Description	Reference
Uplink out-of-synchronization management	The gNodeB maintains uplink timing for a UE in connected mode in real time so that the UE remains in the uplink-synchronized state.	**4.1.4.5 Uplink Out-of-Synchronization Management**
UE inactivity management	After detecting an inactive UE in connected mode, the gNodeB performs inactivity management on the UE to prevent this UE from occupying system resources for a long time.	**4.1.4.6 UE Inactivity Management**
RRC message segmentation	Each RRC message with a size of over 9000 bytes is segmented.	**4.1.4.7 RRC Message Segmentation**

4.1.4.1 RRC Connection Setup

Figure 4-9 shows the RRC connection setup procedure.

Figure 4-9 RRC connection setup procedure



1. A UE sends Msg1 to the gNodeB to initiate contention-based RA.
2. The gNodeB sends an RA response to the UE through Msg2.
3. The UE sends the gNodeB an RRCSetupRequest message, requesting the setup of an RRC connection. (This message is mapped to Msg3 in the random

access procedure. In the RRC connection setup procedure, Msg3 carries the RRCSetupRequest message.) The message carries the RRC connection setup cause and UE identity, in which:

- The RRC connection setup cause is specified by the upper layer.
- The UE identity can be the 5G-S-TMSI or a random number.
  - If the upper layer provides the 5G-S-TMSI, the message contains the 5G-S-TMSI.
  - If no 5G-S-TMSI information is provided, a random number between 0 and (2^{39} – 1) is generated and sent to the gNodeB.

4. The gNodeB sets up UE context.

   If the gNodeB receives multiple RRCSetupRequest messages from the UE within the time window specified by the **gNBConnStateTimer**.***UuMessageWaitingTimer*** parameter, the gNodeB handles only the most recent one.

5. The gNodeB performs the SRB1 admission and resource allocation.

   If the SRB1 admission or resource allocation fails, the gNodeB responds to the UE with an RRCReject message. The RRC connection setup fails. Otherwise, **6** and **7** are performed.

6. The gNodeB responds to the UE with an RRCSetup message, containing SRB1 resource configuration information. (This message is mapped to Msg4 in the random access procedure. In the RRC connection setup procedure, Msg4 carries the RRCSetup message.)

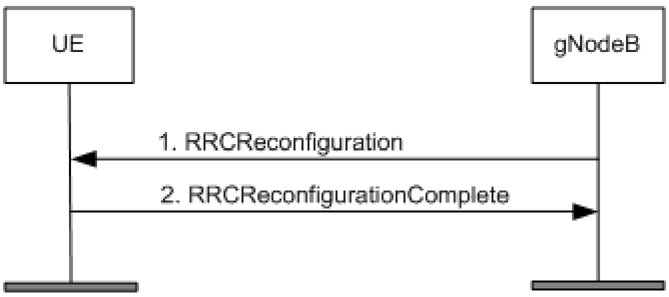
7. The UE configures radio resources based on the SRB1 resource information indicated by the RRCSetup message. It then sends the gNodeB an RRCSetupComplete message indicating that the RRC connection setup is complete.

   After the gNodeB sends the RRCSetup message, it starts a timer to wait for the RRCSetupComplete message. The timer is specified by the **gNBConnStateTimer**.***UuMessageWaitingTimer*** parameter. If the gNodeB does not receive the RRCSetupComplete message from the UE before the timer expires, the RRC connection setup fails.

4.1.4.2 RRC Connection Reconfiguration

Figure 4-10 shows the RRC connection reconfiguration procedure.

Figure 4-10 RRC connection reconfiguration procedure



1. The gNodeB sends an RRCReconfiguration message to a UE to initiate an RRC connection reconfiguration procedure. The RRC connection reconfiguration procedure includes the following:
   – SRB2 setup

     After encryption and integrity protection are complete during UE context setup, the gNodeB sends an RRCReconfiguration message containing the *srb-ToAddModList* IE to the UE, instructing the UE to set up SRB2. For details, see **4.1.5.1 Context Setup**.
   – SRB1/SRB2 modification

     The gNodeB sends an RRCReconfiguration message containing the *srb-ToAddModList* IE to the UE, instructing the UE to modify SRB1 or SRB2. The UE reconfigures the Packet Data Convergence Protocol (PDCP) entity, Radio Link Control (RLC) entity, and dedicated control channel (DCCH) as instructed. SRB1/SRB2 modification can be triggered in many scenarios, such as context modification.
   – DRB setup

     During PDU session setup, the AMF sends a PDU SESSION RESOURCE SETUP REQUEST message to the gNodeB to trigger DRB setup. The gNodeB sends an RRCReconfiguration message containing the *drb-ToAddModList* IE to the UE, instructing the UE to set up a DRB. For details, see **4.1.6.1 PDU Session Setup**.
   – DRB modification

     During PDU session modification, the AMF sends a PDU SESSION RESOURCE MODIFY REQUEST message to the gNodeB to trigger DRB modification. The gNodeB sends an RRCReconfiguration message containing the *drb-ToAddModList* IE to the UE, instructing the UE to modify a DRB. For details, see **4.1.6.2 PDU Session Modification**.
   – DRB release

     During PDU session release, the AMF sends a PDU SESSION RESOURCE RELEASE COMMAND message to the gNodeB to trigger DRB release. The gNodeB sends an RRCReconfiguration message containing the *drb-ToReleaseList* IE to the UE, instructing the UE to release a DRB. For details, see **4.1.6.3 PDU Session Release**.
2. The UE reconfigures radio bearers based on the instructions in the RRCReconfiguration message, and then sends the gNodeB an RRCReconfigurationComplete message indicating that the RRC connection reconfiguration is complete.

4.1.4.3 RRC Connection Release

After an RRC connection is released, the signaling connection between the UE and gNodeB and all the involved radio bearers (SRB1, SRB2, and DRB) are released.

RRC connection release is triggered by context release. For details about RRC connection release, see **4.1.5.3 Context Release**.

4.1.4.4 RRC Connection Reestablishment

RRC connection reestablishment is a service processing procedure initiated by a UE for quick RRC connection reestablishment. The UE can initiate this procedure only when an RRC connection has been set up and the security mode has been enabled. The procedure is triggered in any of the following scenarios:

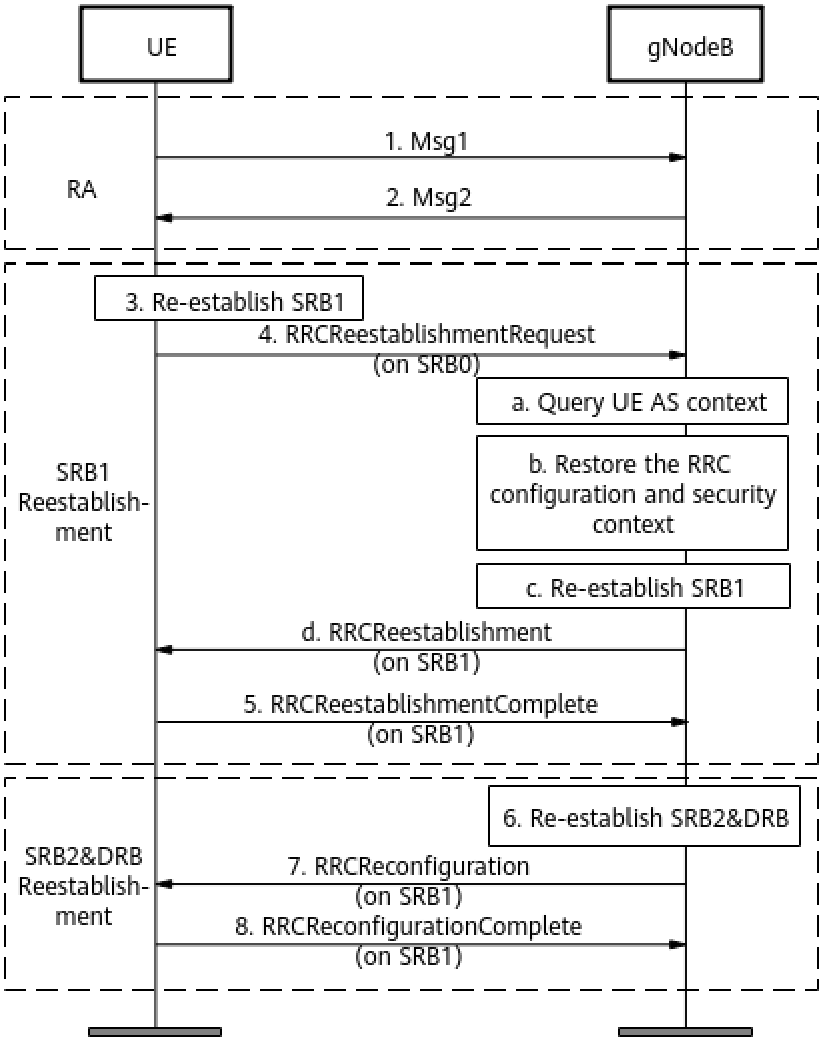
- A radio link failure (RLF) occurs.

  The UE detects an RLF when any of the following conditions is met. For details, see section 7.1.1 "Timers (Informative)" in 3GPP TS 38.331 V15.5.0.

  – The timer specified by the **NRDUCellUeTimerConst**.***T310*** parameter expires.
  – RA fails and the timer specified by the **NRDUCellUeTimerConst**.***T311*** parameter is not running.
  – The number of RLC retransmissions reaches the maximum.
- An inter-RAT outgoing handover fails.
- An intra-RAT handover fails.
- Integrity check fails.
- RRC connection reconfiguration fails.

Figure 4-11 shows the RRC connection reestablishment procedure.

Figure 4-11 RRC connection reestablishment procedure



1. The UE sends Msg1 to the gNodeB to initiate contention-based RA.
2. The gNodeB sends an RA response to the UE through Msg2.
3. The UE reestablishes SRB1 and sends the gNodeB an RRCReestablishmentRequest message carrying the c-RNTI, physCellId, and shortMAC-I used before reestablishment.
4. After receiving the RRCReestablishmentRequest message, the gNodeB performs the following:
   a. Based on the c-RNTI, physCellId, and shortMAC-I, the gNodeB searches for the UE context before RRC connection reestablishment.
   b. The gNodeB restores RRC configuration information and security information according to the UE context.
   c. The gNodeB reestablishes SRB1.
   d. By using SRB1, the gNodeB sends an RRCReestablishment message carrying the *nextHopChainingCount* IE to the UE, instructing the UE to update the AS security key.

   NOTE

   Assume that the gNodeB cannot find the UE context (for example, in inter-gNodeB RRC connection reestablishment scenarios) and receives an RRC connection reestablishment request from the UE:

   - If the **NO_CONTEXT_REEST_SW** option of the **gNodeBParam**.***MobilityOptSwitch*** parameter is selected, the gNodeB finds a neighboring gNodeB that has an Xn connection and serves a cell identified by physCellId. Then, the gNodeB sends a RETRIEVE UE CONTEXT REQUEST message to the neighboring gNodeB to obtain the UE context. For details, see **RRC Connection Reestablishment Without UE Context**.
   - If the **NO_CONTEXT_REEST_SW** option of the **gNodeBParam**.***MobilityOptSwitch*** parameter is deselected, the gNodeB directly performs an RRC connection setup procedure. For details, see **4.1.4.1 RRC Connection Setup**.
5. The UE responds to the gNodeB with an RRCReestablishmentComplete message.
6. The gNodeB continues to reestablish SRB2 and DRB.
7. The gNodeB sends an RRCReconfiguration message to the UE, instructing the UE to reestablish SRB2 and DRB.
8. The UE sends an RRCReconfigurationComplete message to the gNodeB. The RRC connection reestablishment procedure is complete.

RRC Connection Reestablishment Without UE Context

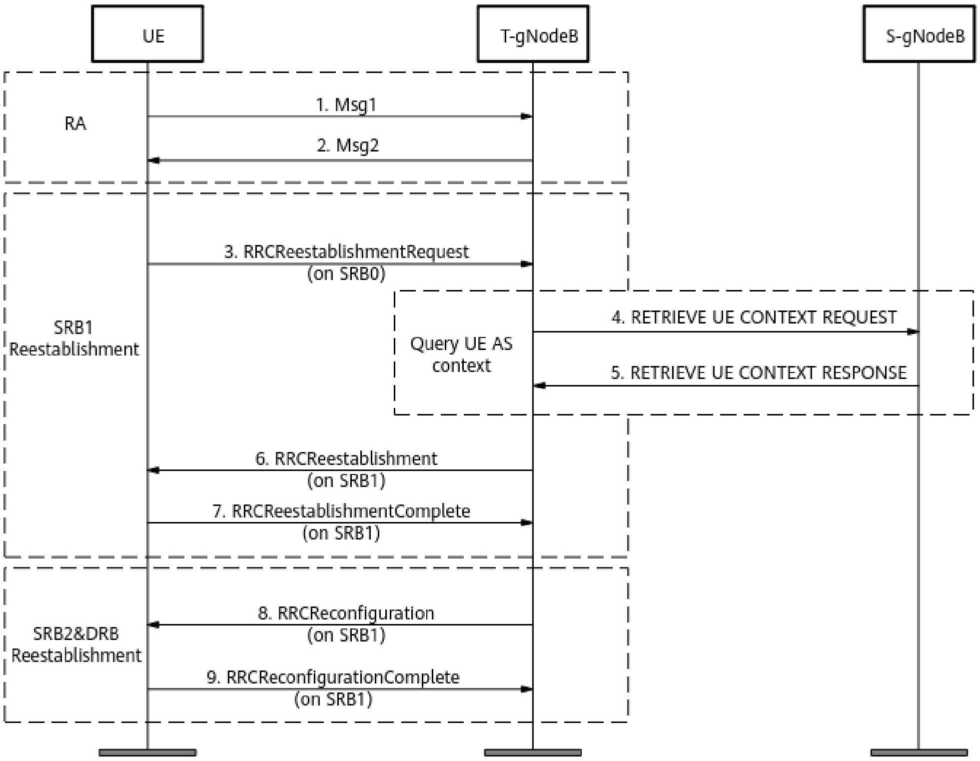
Assume that a UE in connected mode initiates an RRC connection reestablishment request, and the T-gNodeB cannot find the UE context after receiving the request:

- If the **NO_CONTEXT_REEST_SW** option of the **gNodeBParam**.***MobilityOptSwitch*** parameter is selected, an RRC connection reestablishment without UE context is triggered. The T-gNodeB finds an S-gNodeB that has an Xn connection and serves a cell identified by physCellId. The T-gNodeB sends a RETRIEVE UE CONTEXT REQUEST message to the S-gNodeB to obtain the UE context and then completes the RRC connection reestablishment. In this case, the T-gNodeB restores the UE context and continues to provide services for the UE.

- If the **NO_CONTEXT_REEST_SW** option of the **gNodeBParam**.***MobilityOptSwitch*** parameter is deselected, the T-gNodeB performs an RRC connection setup procedure instead of an RRC connection reestablishment without UE context.

Figure 4-12 shows the procedure of RRC connection reestablishment without UE context.

Figure 4-12 Procedure of RRC connection reestablishment without UE context



1. A UE sends Msg1 to the T-gNodeB to initiate contention-based RA.
2. The T-gNodeB sends an RA response to the UE through Msg2.
3. The UE reestablishes SRB1 and sends the T-gNodeB an RRCReestablishmentRequest message carrying the c-RNTI, physCellId, and shortMAC-I used before reestablishment.
4. Since the T-gNodeB cannot find the UE context, it requests the UE context by sending a RETRIEVE UE CONTEXT REQUEST message to the S-gNodeB that serves a cell identified by physCellId.
5. The S-gNodeB searches for the UE context based on the information (including C-RNTI and Failure Cell PCI) carried in the RETRIEVE UE CONTEXT REQUEST message. If the S-gNodeB finds the UE context, it sends the UE context to the T-gNodeB through a RETRIEVE UE CONTEXT RESPONSE message.
6. If the T-gNodeB successfully obtains the UE context from the S-gNodeB, the T-gNodeB sends an RRCReestablishment message to the UE and waits for an RRCReestablishmentComplete message from the UE.
7. The UE responds to the T-gNodeB with an RRCReestablishmentComplete message.

8. The T-gNodeB sends an RRCReconfiguration message to the UE, instructing the UE to reestablish SRB2 and DRB.
9. The UE sends an RRCReconfigurationComplete message to the T-gNodeB. The RRC connection reestablishment procedure is complete.

NOTE

If the T-gNodeB fails to obtain the UE context from the S-gNodeB, the T-gNodeB directly performs an RRC connection setup procedure. For details, see **4.1.4.1 RRC Connection Setup**.

4.1.4.5 Uplink Out-of-Synchronization Management

After the RA is successful, a UE sends a sounding reference signal (SRS) or demodulation reference signal (DMRS) to the gNodeB and the gNodeB performs measurements.

- If the gNodeB can obtain timing alignment information, the gNodeB sends a Timing Advance Command to the UE. The gNodeB and UE use the same uplink time alignment timer (specified by the **NRDUCellUlTaConfig**.***UlTimeAlignmentTimer*** parameter).
  - Before the uplink time alignment timer expires, both the gNodeB and the UE determine that uplink synchronization is achieved. Once a Timing Advance Command is received, the UE restarts the uplink time alignment timer. The gNodeB restarts the uplink time alignment timer upon receiving an ACK to the Timing Advance Command from the UE.
  - A UE cannot achieve uplink synchronization and enters the out-of-synchronization state when the uplink time alignment timer expires in any of the following cases:
    - The UE does not receive a Timing Advance Command before the uplink time alignment timer expires. In this case, the uplink time alignment timer on the UE will expire.
    - The gNodeB does not receive an ACK to the Timing Advance Command from the UE before the uplink time alignment timer expires. In this case, the uplink time alignment timer on the gNodeB will expire.

    In out-of-synchronization state, if the gNodeB or the UE needs to send data, the UE must initiate an RA procedure and restart the uplink time alignment timer.

    When the parameter value of the uplink time alignment timer is set to **INFINITY**, the uplink time alignment timer will never expire.
- If the gNodeB cannot obtain timing alignment information, uplink synchronization cannot be maintained for the UE and out-of-synchronization occurs.

When the UE is in the uplink out-of-synchronization state, the gNodeB instructs the UE to initiate an RA procedure for downlink data transmission or the UE initiates an RA procedure to restore uplink synchronization for uplink data transmission.

4.1.4.6 UE Inactivity Management

After detecting an inactive UE, the gNodeB performs inactivity management on the UE. This prevents the inactive UE from occupying system resources for a long period. A UE becomes inactive when it does not transmit or receive data or when it disconnects from the gNodeB.

When detecting that a UE is in any of the following situations, the gNodeB considers the UE inactive:

- The gNodeB detects that the UE is in the signaling-only connection (no DRB) for a period longer than the value of the **gNBConnStateTimer**.***SigUeNoNasMsgTransTmr*** parameter.
- After the UE sets up a DRB, the gNodeB detects that the UE does not transmit or receive any data (excluding MAC CEs) within the length of the UE inactivity timer (specified by the **NRDUCellQciBearer**.***UeInactivityTimer*** parameter). The UE may set up multiple DRBs, and each DRB corresponds to a QCI. If different UE inactivity timer lengths are configured for these QCIs by setting the **NRDUCellQciBearer**.***UeInactivityTimer*** parameter, the maximum value takes effect.

After considering the UE inactive, the gNodeB initiates a UE state transition or an RRC connection release procedure.

- If the **RRC_INACTIVE_SWITCH** option of the **NRCellAlgoSwitch.*InactiveStrategySwitch*** parameter is selected, the gNodeB instructs the UE to switch from the RRC_CONNECTED state to RRC_INACTIVE state.
- If the **RRC_INACTIVE_SWITCH** option of the **NRCellAlgoSwitch.*InactiveStrategySwitch*** parameter is deselected, the gNodeB sends a UE CONTEXT RELEASE REQUEST message carrying the release cause "User Inactivity" to the AMF. The gNodeB initiates an RRC connection release procedure.

4.1.4.7 RRC Message Segmentation

According to section 4.3.1 "Services provided to upper layers" in 3GPP TS 38.323 V17.0.0, the maximum size of a PDCP SDU is 9000 bytes. Therefore, the size of an RRC message carried by a PDCP SDU cannot exceed 9000 bytes. In compliance with the protocol, an RRC message with a size of more than 9000 bytes can be segmented to resolve the issue that no PDCP SDU can accommodate such long messages. This function is controlled by the **RRC_MESSAGE_SEGMENT_SW** option of the **gNodeBParam**.***ProcessSwitch*** parameter. For details about RRC message segmentation, see sections 5.7.6 "DL message segment transfer" and 5.7.7 "UL message segment transfer" in 3GPP TS 38.331 V16.7.0.

Table 4-6 provides details about RRC message segmentation procedures.

Table 4-6 RRC message segmentation procedures

Type	Prerequisite	RRC Message Size	Procedure
Uplink RRC message segmentation	The gNodeB includes the *rrc-SegAllowed-r16* IE in the UECapabilityEnquiry message sent to the UE. The value of this IE is "enabled".	The UECapabilityInformation message exceeds 9000 bytes.	The UE encodes and segments this message so that each segment does not exceed 9000 bytes. The UE then uses ULDedicatedMessageSegment messages to convey these segments, reporting its capabilities to the gNodeB.
		The UECapabilityInformation message does not exceed 9000 bytes.	This message does not need to be segmented. The UE reports its capabilities through the UECapabilityInformation message.
Downlink RRC message segmentation	A UE capability message contains the *dl-DedicatedMessageSegmentation-r16* IE whose value is "supported".	The RRCResume or RRCReconfiguration message exceeds 9000 bytes.	The gNodeB encodes and segments this message so that each segment does not exceed 9000 bytes. The gNodeB then uses DLDedicatedMessageSegment messages to convey these segments to the UE.
		The RRCResume or RRCReconfiguration message does not exceed 9000 bytes.	This message does not need to be segmented.

4.1.5 Context Management

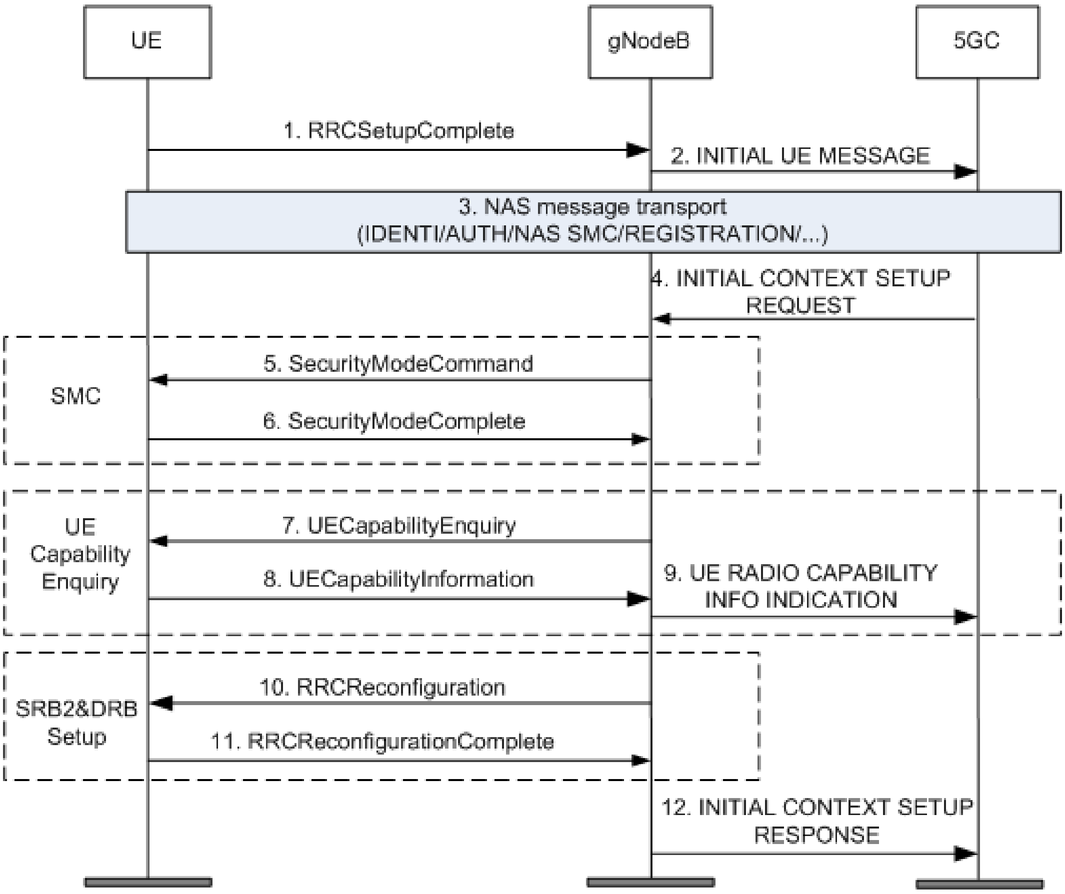
Context management includes context setup, context modification, and context release. For details, see **4.1.5.1 Context Setup**, **4.1.5.2 Context Modification**, and **4.1.5.3 Context Release**. For more details about context management, see section 8.3 "UE Context Management Procedures" in 3GPP TS 38.413 V15.5.0.

4.1.5.1 Context Setup

After the RRC connection is set up, the gNodeB sends an INITIAL UE MESSAGE to trigger NG-C connection setup and initial context setup procedures. The message contains the PDU session, security key, handover restriction list, UE radio capability, and security capability.

Figure 4-13 shows the initial context setup procedure.

Figure 4-13 Context setup procedure



1. After the RRC connection is set up, the UE sends an RRCSetupComplete message to the gNodeB. The RRCSetupComplete message contains the selectedPLMN-Identity, registeredAMF, s-nssai-list, and NAS message.
2. The gNodeB allocates a dedicated RAN-UE-NGAP-ID to the UE, and selects an AMF node based on the selectedPLMN-Identity, registeredAMF, and s-nssai-list. Then, it sends the NAS message carried in the RRCSetupComplete message to the AMF through an INITIAL UE MESSAGE, triggering an NG-C connection setup procedure.
3. The gNodeB transparently transmits the NAS direct transfer messages between the UE and AMF to complete the identity query, authentication, NAS security mode, and registration.
4. The AMF sends an INITIAL CONTEXT SETUP REQUEST message to the gNodeB, triggering an initial context setup procedure.

NOTE

- Only when the INITIAL CONTEXT SETUP REQUEST message does not carry the *UE Radio Capability* IE, the gNodeB sends a UECapabilityEnquiry message to the UE after security mode procedure completion to initiate a UE capability query procedure, as indicated by **7** to **9**. Otherwise, **7** to **9** are skipped.
- Only when the INITIAL CONTEXT SETUP REQUEST message carries the *PDU Session Resource Setup Request List* IE, the gNodeB sends an RRCReconfiguration message with encryption and integrity protection implemented to the UE after UE capability query completion, instructing the UE to set up SRB2 and a DRB, as indicated by **10** and **11**. Otherwise, **10** and **11** are skipped.

5. The gNodeB sends a SecurityModeCommand message to the UE, instructing the UE to start integrity protection and encryption. Then, downlink encryption starts.
6. Based on the integrity protection and encryption algorithms indicated by the SecurityModeCommand message, the UE derives a key and sends a SecurityModeComplete message to the gNodeB. Then, uplink encryption starts.
7. The gNodeB sends a UECapabilityEnquiry message to the UE to initiate a UE capability query procedure.
8. The UE sends a UECapabilityInformation message carrying the UE capability information to the gNodeB.
9. The gNodeB transparently transmits the UE capability to the AMF through a UE RADIO CAPABILITY INFO INDICATION message.
10. The gNodeB sends an RRCReconfiguration message to the UE, instructing the UE to set up SRB2 and a DRB.

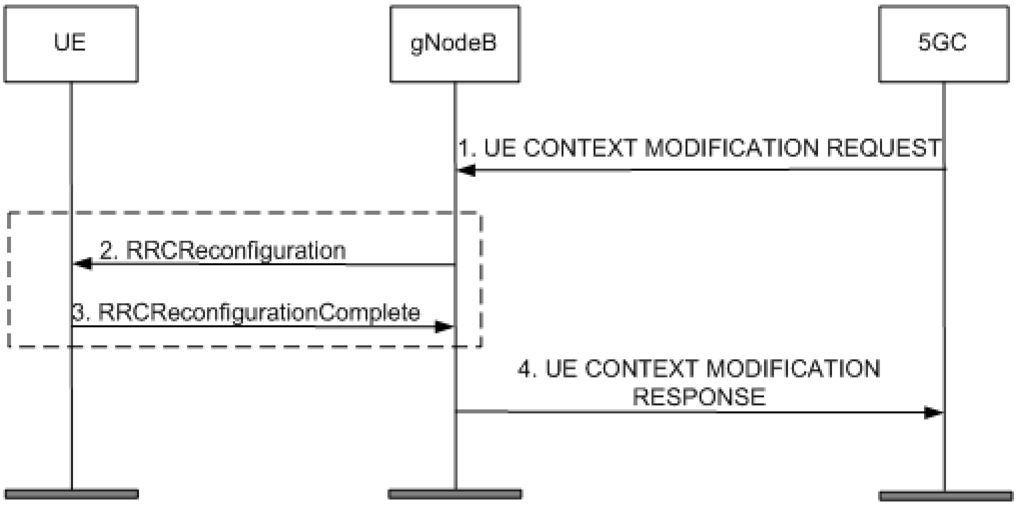
    After encryption and integrity protection are complete during dedicated NG-C connection setup, the gNodeB sends an RRCReconfiguration message containing the *srb-ToAddModList* IE to the UE, instructing the UE to set up SRB2 and a DRB.
11. After receiving the RRCReconfiguration message, the UE starts the setup of SRB2 and a DRB. The UE performs the following operations as instructed:
    - Sets up a PDCP entity and configures related security parameters.
    - Sets up and configures an RLC entity.
    - Sets up and configures a DCCH.

    After SRB2 and a DRB are set up, the UE sends an RRCReconfigurationComplete message to the gNodeB.
12. The gNodeB sends an INITIAL CONTEXT SETUP RESPONSE message to the AMF.

4.1.5.2 Context Modification

Figure 4-14 shows the context modification procedure.

Figure 4-14 Context modification procedure



1. The AMF sends a UE CONTEXT MODIFICATION REQUEST to the gNodeB, triggering a UE context modification procedure.

   NOTE

   When the UE CONTEXT MODIFICATION REQUEST message contains the *Security Key* IE, the gNodeB triggers a key update procedure. During the key update procedure, the gNodeB initiates an RRC connection reconfiguration procedure and the UE reconfigures radio bearers, as indicated by **2** and **3**. Otherwise, **2** and **3** are skipped.

2. After deriving a key based on the *Security Key* IE, the gNodeB sends an RRCReconfiguration message to the UE, instructing the UE to update the key. For details about key derivation, see 3GPP TS 33.501 V15.5.0.
3. After updating the key, the UE sends an RRCReconfigurationComplete message to the gNodeB.
4. The gNodeB sends the AMF a UE CONTEXT MODIFICATION RESPONSE message indicating that the UE context modification is complete.

4.1.5.3 Context Release

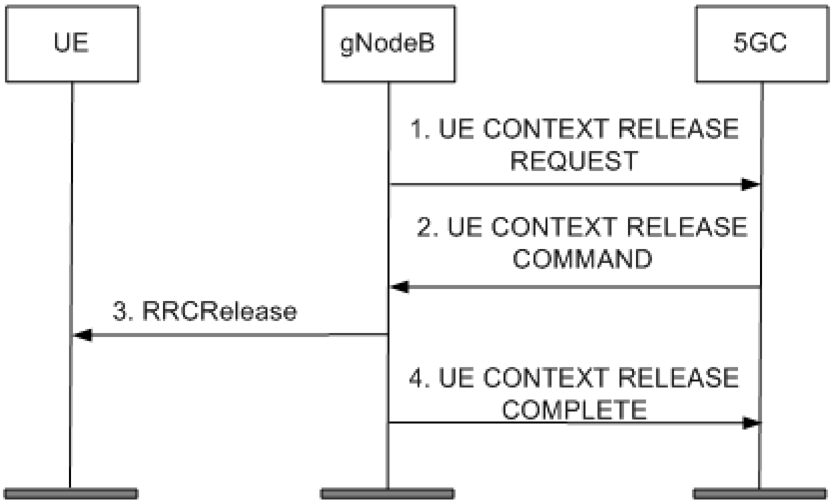
Context release involves releasing the signaling connection between the gNodeB and 5GC. A context release procedure can be initiated by the gNodeB or the AMF.

- gNodeB-triggered context release: The gNodeB sends a UE CONTEXT RELEASE REQUEST message to the AMF upon detecting a UE connection exception, such as the UE inactivity.
- AMF-triggered context release: The AMF sends a UE CONTEXT RELEASE COMMAND message to the gNodeB when the AMF decides to abort the service of a UE or a UE decides to abort the service and informs the AMF through NAS signaling.

In scenarios with only the context of a UE but no PDU sessions, the UE may automatically release the context after a period. However, the network does not detect the release. When such an exception occurs, the gNodeB resources are occupied for an extended duration. To avoid this issue, a protection measure is introduced. Specifically, the gNodeB monitors whether NAS signaling is exchanged between the UE and the AMF within the length of a timer, and if not, triggers a context release. The timer is specified by the **gNBConnStateTimer**.***SigUeNoNasMsgTransTmr*** parameter.

Figure 4-15 shows the context release procedure.

Figure 4-15 Context release procedure



1. The gNodeB sends a UE CONTEXT RELEASE REQUEST message to the AMF, requesting a context release.

   NOTE

   1 is involved only when the context release is triggered by the gNodeB.

2. The AMF sends a UE CONTEXT RELEASE COMMAND message to the gNodeB, instructing the gNodeB to release the UE context.
3. The gNodeB sends an RRCRelease message to the UE, instructing the UE to release the RRC connection.
4. The gNodeB sends the AMF a UE CONTEXT RELEASE COMPLETE message indicating that the context release is complete.

   After receiving the UE CONTEXT RELEASE COMPLETE message, the AMF releases the NAS context information and AS context information corresponding to the UE. Then, the UE switches from connected mode to idle mode.

Compatibility Optimization for the UE Release Procedure Upon UE Inactivity

During a context release procedure, if the gNodeB releases the context without processing the downlink signaling messages (except the UE CONTEXT RELEASE COMMAND message) delivered by the AMF, possible conflicts in the procedure of UE releases due to UE inactivity will increase the number of QoS flow setup failures. The following are typical downlink signaling messages that are not processed:

- PDU SESSION RESOURCE SETUP REQUEST
- PDU SESSION RESOURCE RELEASE COMMAND
- PDU SESSION RESOURCE MODIFY REQUEST

- UE CONTEXT MODIFICATION REQUEST
- DOWNLINK NAS TRANSPORT
- LOCATION REPORTING CONTROL

Against this backdrop, compatibility optimization for the UE release procedure upon UE inactivity is introduced. With this function, after the gNodeB sends a UE CONTEXT RELEASE REQUEST message due to UE inactivity and receives a downlink signaling message from the AMF, the gNodeB processes the downlink signaling message and does not release the context. This reduces the number of QoS flow setup failures that are caused by conflicts in the procedure of UE releases due to UE inactivity. Compatibility optimization for the UE release procedure upon UE inactivity is controlled by the **UE_INACT_REL_PROC_COMPT_OPT_SW** option of the **gNodeBParam.*CompatibilityAlgoSwitch*** parameter. To learn more about this function, see section 8.3.2 "UE Context Release Request (NG-RAN node initiated)" in 3GPP TS 38.413 V18.2.0.

4.1.6 PDU Session Management

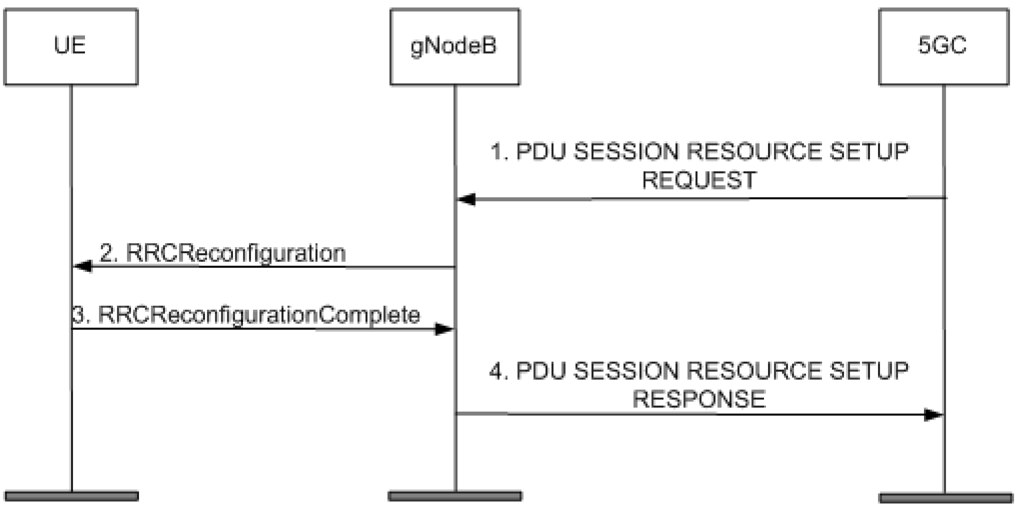
PDU session management involves the following:

- PDU session setup: a procedure for setting up DRBs and NG-U transmission tunnels for QoS flows corresponding to one or more PDU sessions. For details, see **4.1.6.1 PDU Session Setup**.
- PDU session modification: a procedure for modifying DRBs and NG-U transmission tunnels for QoS flows corresponding to one or more PDU sessions. For details, see **4.1.6.2 PDU Session Modification**.
- PDU session release: a procedure for releasing DRBs and NG-U transmission tunnels for QoS flows corresponding to one or more PDU sessions. For details, see **4.1.6.3 PDU Session Release**.

4.1.6.1 PDU Session Setup

Figure 4-16 shows the PDU session setup procedure.

Figure 4-16 PDU session setup procedure



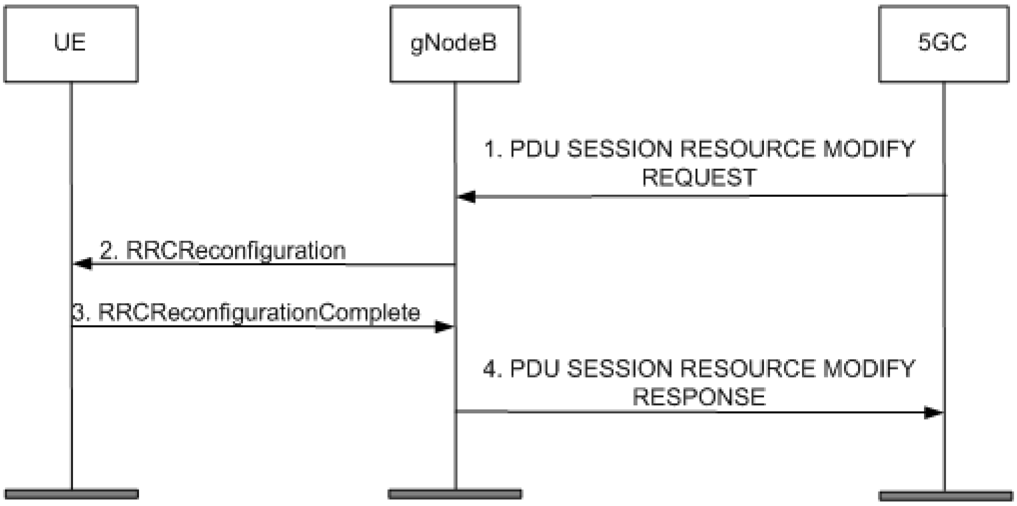
1. The AMF sends a PDU SESSION RESOURCE SETUP REQUEST message to the gNodeB. The message contains a list of PDU sessions to be set up, a list of QoS flows of each PDU session, and the quality attribute of each QoS flow.
2. The gNodeB maps QoS flows to DRBs based on the QoS flow quality attributes and MML-configured policy. It then sends an RRCReconfiguration message to the UE, instructing the UE to set up DRBs.
3. Based on the *drb-ToAddModList* IE contained in the RRCReconfiguration message, the UE sets up DRBs. The UE performs the following operations as instructed:
   - Sets up a PDCP entity and configures related security parameters.
   - Sets up and configures an RLC entity.
   - Sets up and configures a dedicated traffic channel (DTCH).

   After setting up DRBs, the UE sends an RRCReconfigurationComplete message to the gNodeB.
4. The gNodeB sends the AMF a PDU SESSION RESOURCE SETUP RESPONSE message indicating that the PDU session setup is complete.

4.1.6.2 PDU Session Modification

Figure 4-17 shows the PDU session modification procedure.

Figure 4-17 PDU session modification procedure



1. The AMF sends a PDU SESSION RESOURCE MODIFY REQUEST message to the gNodeB. The message contains information including a QoS Flow Add or Modify Response List and a QoS Flow to Release List.
2. The gNodeB modifies DRBs based on the QoS policy and sends the UE an RRCReconfiguration message carrying the *drb-ToAddModList* IE. DRB modification is performed in the following three scenarios:
   - Adding a DRB: New QoS flows cannot be mapped to the existing DRBs. And a new DRB needs to be added to meet the QoS requirements.
   - Deleting a DRB: If all QoS flows mapped to a DRB have been deleted, this DRB needs to be deleted.

- Modifying a DRB: A QoS flow mapping needs to be added to or deleted from an existing DRB.

3. The UE reconfigures the PDCP entity, RLC entity, and DTCH according to the instructions in the RRCReconfiguration message. After reconfiguration, the UE sends an RRCReconfigurationComplete message to the gNodeB.
4. The gNodeB sends the AMF a PDU SESSION RESOURCE MODIFY RESPONSE message indicating that the PDU session modification is complete.

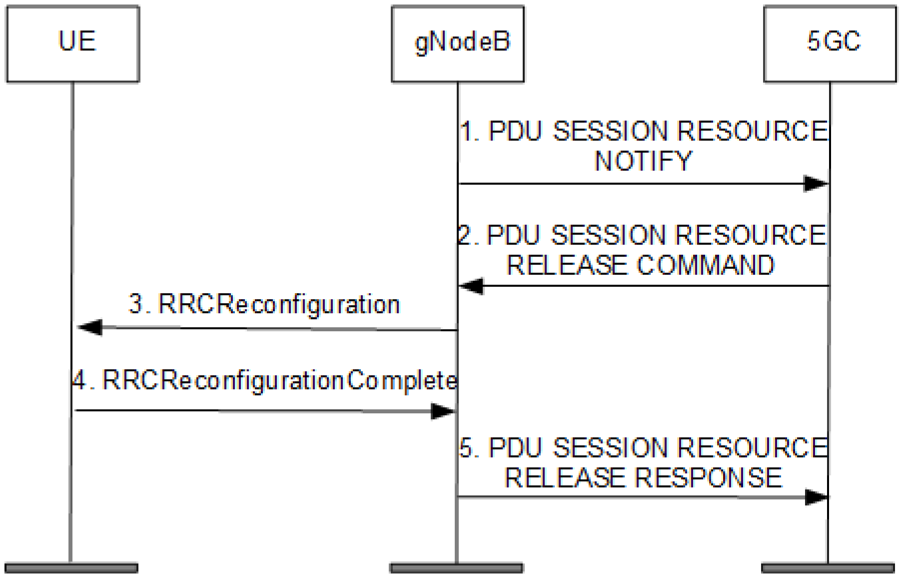
4.1.6.3 PDU Session Release

The gNodeB or the AMF can initiate a PDU session release procedure.

- gNodeB-triggered PDU session release:
  - The gNodeB reallocates a new NG-U address upon detecting an NG-U transmission fault. If the allocation fails, the gNodeB sends a PDU SESSION RESOURCE NOTIFY message to the AMF, requesting the AMF to initiate a PDU session release procedure.
  - When a GTP-U error occurs on the NG-U path, the gNodeB sends the AMF a PDU SESSION RESOURCE NOTIFY message that contains the *User Plane Error Indicator* IE with the value of "GTP-U Error Indication Received". In consequence, the AMF initiates a PDU session release procedure.
  - If the QoS flow guaranteed bit rate (GBR) cannot be provided, the gNodeB sends a PDU SESSION RESOURCE NOTIFY message to the AMF, requesting the AMF to initiate a PDU session release procedure.
- AMF-triggered PDU session release:
  - The AMF sends a PDU SESSION RESOURCE RELEASE COMMAND message to the gNodeB to trigger a PDU session release procedure when the AMF decides to abort the service of the UE.
  - The AMF sends a PDU SESSION RESOURCE RELEASE COMMAND message to the gNodeB to trigger a PDU session release procedure when the UE decides to abort the service and informs the AMF through NAS signaling.

Figure 4-18 shows the PDU session release procedure.

Figure 4-18 PDU session release procedure



1. The gNodeB sends a PDU SESSION RESOURCE NOTIFY message to the AMF, requesting the AMF to trigger a PDU session release procedure.

   NOTE

   1 is involved only when the PDU session release is triggered by the gNodeB.
2. The AMF sends the gNodeB a PDU SESSION RESOURCE RELEASE COMMAND message carrying a list of PDU sessions to be released.
3. The gNodeB sends an RRCReconfiguration message to the UE, instructing the UE to release PDU sessions.
4. Based on the *drb-ToReleaseList* IE contained in the RRCReconfiguration message, the UE releases all of the resources related to the DRBs as instructed. After the PDU session release is complete, the UE sends an RRCReconfigurationComplete message to the gNodeB.
5. The gNodeB deletes the corresponding DRBs and NG-U transmission tunnels, and then sends a PDU SESSION RESOURCE RELEASE RESPONSE message to the AMF.

4.2 Network Analysis

4.2.1 Benefits

If the **NO_CONTEXT_REEST_SW** option of the **gNodeBParam**.***MobilityOptSwitch*** parameter is selected, the number of RRC connection reestablishments increases and the number of RRC connection setups decreases.

If the **PAGING_INTRF_RANDOM_SW** option of the **NRDUCellPagingConfig**.***PagingSchAlgoSwitch*** parameter is selected, the interference from neighboring cells to UEs in the local cell decreases and the paging success rate increases.

OSI payload reduction reduces the number of bits conveyed by OSI and therefore reduces the number of RBs used for each OSI transmission, when power aggregation takes effect in a high-frequency TDD cell in SA networking or SA and NSA hybrid networking. This way, it improves cell coverage.

If the paging capacity mode of a cell is enhanced mode 1 (that is, the **NRDUCellPagingConfig**.***PagingCapacityMode*** parameter is set to **ENHANCE_MODE_1**), the paging capacity of the cell increases.

Compatibility optimization for the UE release procedure upon UE inactivity reduces the number of QoS flow setup failures that are caused by conflicts in the procedure of UE releases due to UE inactivity.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

If the **NRDUCellPagingConfig.*MaxPagingMsgNum*** parameter is set to **32**, there is no impact on the network.

If the **NRDUCellPagingConfig.*MaxPagingMsgNum*** parameter is set to a value less than 32:

- In light-load scenarios, the paging success rate increases. The larger the parameter value, the lower the probability of congestion by paging messages.
- In heavy-load scenarios with paging message congestion, **N.Paging.UU.Ue.Num** decreases, **N.Paging.Dis.PchCong** increases, the number of online UEs in the cell and the number of RBs occupied by paging messages decrease, and the downlink cell throughput and user-perceived rate rise. It is recommended that this parameter be set to a value greater than or equal to 4 in heavy-load scenarios.
- When radio resources are sufficient, the maximum number of paging messages supported within a paging occasion slightly changes.

If the **NRDUCellPagingConfig**.***PagingTransPolicy*** parameter is set to **PRI_DIFFERENTIATED**, the paging delay of UEs with high paging priorities may decrease and that of UEs with low paging priorities may increase in the case of a high paging load.

A smaller value of the **NRDUCellPagingConfig**.***PagingFrameTransPolicy*** parameter results in more paging occasions but more occupied time-frequency resources, decreasing the average cell rate. A larger value of this parameter results in fewer occupied time-frequency resources but fewer paging occasions, decreasing the paging success rate.

If the **PAGING_CR_ADAP_SW** option of the **NRDUCellPagingConfig.*PagingSchAlgoSwitch*** parameter is selected:

- The paging success rate increases (especially for voice service UEs), and the probability of missing calls decreases.
- As more PDSCH RBs are used for paging message scheduling, fewer RBs are available for data services, decreasing the downlink cell throughput and downlink user-perceived rate.

RRC message segmentation allows an RRC message with a size of more than 9000 bytes to be sent over the Uu interface. The segmentation of long messages increases the number of transmissions and transmission delay over the Uu interface.

If the **PAGING_INTRF_RANDOM_SW** option of the **NRDUCellPagingConfig**.***PagingSchAlgoSwitch*** parameter is selected, then the available bandwidth for paging message scheduling decreases, the number of scheduling times may increase, the CCE usage increases, and the downlink cell throughput and user-perceived rate decrease. In addition, the UE access delay increases, and therefore the access success rate decreases.

In SA networking or SA and NSA hybrid networking, OSI payload reduction slightly decreases the **User Downlink Average Throughput**. In other scenarios, this function has no impact.

After the **NRDUCellPagingConfig**.***DefaultPagingCycle*** parameter is set to a smaller value, the success rates of access, handovers, RRC setup, RRC reestablishment, and RRC resume may decrease.

If the paging capacity mode of a cell is enhanced mode 1 (that is, the **NRDUCellPagingConfig**.***PagingCapacityMode*** parameter is set to **ENHANCE_MODE_1**), the number of SSB beams may decrease and the SSB beam coverage capability deteriorates. As a result, the following performance indicators may change: average uplink/downlink UE throughput, uplink/downlink cell traffic volume, average uplink/downlink cell throughput, UE handover delay, the number of UEs, service drops, accessibility, intra-/inter-base-station handover success rate, uplink/downlink RLC-layer indicators of UEs, first-packet delay, uplink/downlink MU-MIMO pairing indicators, CCE aggregation level indicators, average uplink/downlink rank, average uplink/downlink MCS index, average uplink/downlink PRB usage, average uplink/downlink number of scheduled UEs, average uplink/downlink CQI, uplink/downlink BLER, uplink/downlink CPU usage, uplink/downlink MAC-layer processing delay, uplink/downlink RLC-layer processing delay, and RAT-specific base station energy consumption. In addition, within three hours after the paging capacity mode is changed, UEs may not receive paging messages due to UE incompatibility.

The paging origin function increases the number of bits carried by a paging message over the Uu interface. It may increase the code rate of paging messages and therefore worsens the paging coverage performance.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Initial CCE aggregation level selection optimization	**AGG_LVL_INIT_SELECT_OPT_SW** option of the **NRDUCellPdcch.*PdcchAlgoEnhSwitch*** parameter	*Channel Management*	The initial CCE aggregation level selection optimization function must be enabled before the full-bandwidth initial BWP configuration function (controlled by the **INIT_BWP_FULL_BW_SW** option of the **NRDUCellAlgoSwitch**.***BwpConfigPolicySwitch*** parameter) and PDCCH rate matching function (controlled by the **PDCCH_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter) are both enabled.
Low-frequency TDD	Paging capacity enhancement	**PAGING_CAPACITY_ENH_SW** option of the **NRDUCellPagingConfig.*PagingCapacityMode*** parameter	*Flow Control*	Setting the **NRDUCellPagingConfig**.***PagingCapacityMode*** parameter to **ENHANCE_MODE_1** requires paging capacity enhancement to be enabled.
Low-frequency TDD	Deployment Position	**NRDUCell.*DeployPosition***	None	The **NRDUCellPagingConfig**.***PagingCapacityMode*** parameter can be set to **ENHANCE_MODE_1** only when the **NRDUCell.*DeployPosition*** parameter is set to **DEPLOY_INTEGRATED**.

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Allocation optimization for common control information	**COMM_CTRL_INFO_ALLOC_OPT_SW** option of the **NRDUCellPdsch**.***DlPdschAlgoSwitch*** parameter	None	Paging interference randomization does not work with allocation optimization for common control information. The two functions cannot be enabled at the same time.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Remote interference adaptive avoidance	**RIM_ADAPT_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	If remote interference adaptive avoidance is enabled, the **NRDUCellPagingConfig**.***PagingCapacityMode*** parameter can only be set to **DEFAULT_MODE**.
Low-frequency TDD	Enhanced remote interference adaptive avoidance	**RIM_ADAPT_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch**.***RimAlgoSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	If enhanced remote interference adaptive avoidance is enabled, the **NRDUCellPagingConfig**.***PagingCapacityMode*** parameter can only be set to **DEFAULT_MODE**.
Low-frequency TDD	Paging-OSI staggering	**PAGING_OSI_STAGGER_SW** option of the **NRDUCellPagingConfig**.***PagingSchAlgoSwitch*** parameter	None	The paging-OSI staggering switch cannot be turned on when the **NRDUCellPagingConfig**.***PagingCapacityMode*** parameter is set to **ENHANCE_MODE_1**.
Low-frequency TDD	SSB Time Position	**NRDUCell**.***SsbTimePos***	None	**NRDUCellPagingConfig**.***PagingCapacityMode*** can be set to **ENHANCE_MODE_1** only when **SSB Time Position** is set to **DEFAULT**.
Low-frequency TDD	PEI	**PEI_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter	*UE Power Saving*	The PEI function cannot be enabled when the **NRDUCellPagingConfig**.***PagingCapacityMode*** parameter is set to **ENHANCE_MODE_1**.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	eDRX in idle mode	**IDLE_MODE_EDRX_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter	*RedCap*	eDRX in idle mode cannot be enabled when the **NRDUCellPagingConfig**.***PagingCapacityMode*** parameter is set to **ENHANCE_MODE_1**.
Low-frequency TDD	DL Blanking	**NRDUCell**.***PeerCellSlotAssignment***	*Standards Compliance*	DL Blanking cannot be enabled when the **NRDUCellPagingConfig**.***PagingCapacityMode*** parameter is set to **ENHANCE_MODE_1**.

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag***	*High Speed Mobility*	UEs are more likely to be out of synchronization in high-speed mobility scenarios. The longer the uplink time alignment timer, the higher the probability that UEs are out of synchronization.
Low-frequency TDD	Low latency and high reliability	**HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	The full-bandwidth initial BWP is recommended when uplink interference randomization-based scheduling is enabled for fixed resources of high-reliability services.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL CoMP	**INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	When DL CoMP and paging code rate adaptation are both enabled, the gNodeB uses the target paging code rate defined by the adaptation function to predict the amount of frequency-domain PDSCH resources occupied by paging messages. The predicted amount is greater than when paging code rate adaptation does not take effect. As a result, the cell has fewer available PDSCH resources remaining.
Low-frequency TDD	Inter-gNodeB CA	**INTER_GNODEB_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*CA Experience Improvement*	When inter-gNodeB CA and paging code rate adaptation are both enabled, the gNodeB uses the target paging code rate defined by the adaptation function to predict the amount of frequency-domain PDSCH resources occupied by paging messages. The predicted amount is greater than when paging code rate adaptation does not take effect. As a result, the cell has fewer available PDSCH resources remaining.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode***	*Cell Combination*	When Cell Combination and paging code rate adaptation are both enabled, the gNodeB uses the target paging code rate defined by the adaptation function to predict the amount of frequency-domain PDSCH resources occupied by paging messages. The predicted amount is greater than when paging code rate adaptation does not take effect. As a result, the cell has fewer available PDSCH resources remaining.
Low-frequency TDD	Coordinated interference management	**INTRA_GNB_DL_CS_CBF_SW** and **INTER_GNB_DL_CS_CBF_SW** options of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*Coordinated Interference Management (Low-Frequency TDD)*	When coordinated interference management and paging code rate adaptation are both enabled, the gNodeB uses the target paging code rate defined by the adaptation function to predict the amount of frequency-domain PDSCH resources occupied by paging messages. The predicted amount is greater than when paging code rate adaptation does not take effect. As a result, the cell has fewer available PDSCH resources remaining.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	This function is enabled if both of the following options are selected: LTE: **LTE_NR_DYN_POWER_SHARING_SW** option of the **CellDynPowerSharing**.***DynamicPowerSharingSwitch*** parameter NR: **LTE_NR_DYN_POWER_SHARING_SW** option of the **NRDUCellAlgoSwitch**.***DynPowerSharingSwitch*** parameter	*LTE and NR Power Allocation*	When LTE TDD and NR Flash Dynamic Power Sharing and paging code rate adaptation are both enabled, the gNodeB uses the target paging code rate defined by the adaptation function to predict the amount of frequency-domain PDSCH resources occupied by paging messages. The predicted amount is greater than when paging code rate adaptation does not take effect. As a result, the cell has fewer available PDSCH resources remaining.
High-frequency TDD	DRX	**BASIC_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter	*DRX*	The gNodeB does not schedule SRSs in DRX-defined sleep time. Therefore, the gNodeB will receive fewer valid SRSs from UEs, which may cause inaccurate uplink timing and increase the uplink BLER.
Low-frequency TDD	RF channel intelligent shutdown	**RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	After RF channel intelligent shutdown takes effect, beam weights may change, resulting in failures of UEs to receive paging messages.
Low-frequency TDD	Low-power RF channel dynamic shutdown	**LOW_POWER_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	After low-power RF channel dynamic shutdown takes effect, beam weights may change, resulting in failures of UEs to receive paging messages.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TTI-level channel shutdown	**TTI_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	After TTI-level channel shutdown takes effect, beam coverage areas alter. As a result, UEs may fail to receive paging messages.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Cells

Dynamic adjustment of the number of PFs does not work in SDL cells, whose **NRDUCell**.***DuplexMode*** is set to **CELL_SDL**.

(TDD) The **NRDUCellPagingConfig**.***PagingCapacityMode*** parameter can be set to **ENHANCE_MODE_1** for cells with the following slot configurations and slot structures:

- 4:1 slot configuration (**NRDUCell**.***SlotAssignment*** set to **4_1_DDDSU**) with slot structure (specified by **NRDUCell**.***SlotStructure***) **SS1**, **SS2**, **SS3**, **SS4**, **SS5**, or **SS6**
- Single-period 8:2 slot configuration (**NRDUCell**.***SlotAssignment*** set to **8_2_DDDDDDDSUU**) with slot structure (specified by **NRDUCell**.***SlotStructure***) **SS51**, **SS52**, **SS53**, **SS54**, **SS55**, or **SS56**

- Dual-period 8:2 slot configuration (**NRDUCell**.***SlotAssignment*** set to **8_2_DDDSUUDDDD**) with slot structure (specified by **NRDUCell**.***SlotStructure***) **SS81**, **SS82**, **SS83**, **SS84**, **SS85**, or **SS86**

4.4 Operation and Maintenance

The basic SA networking functions described in this chapter are enabled by default along with cell setup. For details about cell setup in SA networking, see *Cell Management*.

After a cell is set up, you can determine whether to enable the functions described in **4.4.1.1 Data Preparation** based on site conditions.

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-7 describes the parameters used for activation of user-activatable functions after cell setup.

Table 4-7 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Downlink Schedule Algorithm Switch	**NRDUCellDlSch**.***DlSchAlgoSwitch***	To extend the coverage area of a high-frequency TDD cell with power aggregation enabled, you are advised to select the **OSI_PAYLOAD_REDUCE_SW** option of this parameter.
Mobility Optimization Switch	**gNodeBParam**.***MobilityOptSwitch***	To enable RRC connection reestablishment without UE context, select the **NO_CONTEXT_REEST_SW** option.
Maximum Paging Message Number	**NRDUCellPagingConfig**.***MaxPagingMsgNum***	Set this parameter based on the number of paging messages and the paging success rate.
Power Saving Switch	**NRDUCellAlgoSwitch**.***PowerSavingSwitch***	To enable dynamic adjustment of the number of PFs, select the **PAGING_FRAME_NUM_DYN_ADJ_SW** option.

Parameter Name	Parameter ID	Setting Notes
Paging Scheduling Algorithm Switch	**NRDUCellPagingConfig**.*PagingSchAlgoSwitch*	• To enable paging coding rate adaptation, select the **PAGING_CR_ADAP_SW** option. • To enable paging interference randomization, select the **PAGING_INTRF_RANDOM_SW** option.
Process Switch	**gNodeBParam**.*ProcessSwitch*	To enable RRC message segmentation, select the **RRC_MESSAGE_SEGMENT_SW** option.
Paging Capacity Mode	**NRDUCellPagingConfig**.*PagingCapacityMode*	Set this parameter based on the network plan. The setting of **ENHANCE_MODE_1** is not recommended for common scenarios. For sites with favorable cell coverage and high requirements for paging capacity, this parameter can be set to **ENHANCE_MODE_1** as required after thorough evaluation of the impact. Due to UE compatibility issues, it is recommended that this parameter be reconfigured only in the early morning, when there are few online UEs.
Compatibility Algorithm Switch	**gNodeBParam**.*CompatibilityAlgoSwitch*	To enable compatibility optimization for the UE release procedure upon UE inactivity, select the **UE_INACT_REL_PROC_COMPT_OPT_SW** option.
Paging Algorithm Switch	**gNodeBAlgo**.*PagingAlgoSwitch*	To enable the paging origin function, select the **PAGING_ORIGIN_SW** option.
Paging Frame Transmission Policy	**NRDUCellPagingConfig**.*PagingFrameTransPolicy*	Use the default value.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling OSI payload reduction
MOD NRDUCELLDLSCH: NrDuCellId=0, DlSchAlgoSwitch=OSI_PAYLOAD_REDUCE_SW-1;
//Enabling RRC message segmentation
MOD GNODEBPARAM: ProcessSwitch=RRC_MESSAGE_SEGMENT_SW-1;
//Enabling RRC connection reestablishment without UE context
MOD GNODEBPARAM: MobilityOptSwitch=NO_CONTEXT_REEST_SW-1;
//Enabling paging scheduling
MOD NRDUCELLPAGINGCONFIG: NrDuCellId=0, PagingSchAlgoSwitch=PAGING_CR_ADAP_SW-1;
//Configuring the paging frame transmission policy
MOD NRDUCELLPAGINGCONFIG: NrDuCellId=0, PagingFrameTransPolicy=0;
//Setting the maximum number of paging messages to 30
MOD NRDUCELLPAGINGCONFIG: NrDuCellId=0, MaxPagingMsgNum=30;
//Enabling paging interference randomization
MOD NRDUCELLPAGINGCONFIG: NrDuCellId=0, PagingSchAlgoSwitch=PAGING_INTRF_RANDOM_SW-1;
//Enabling dynamic adjustment of the number of PFs
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=PAGING_FRAME_NUM_DYN_ADJ_SW-1;
//Setting the paging capacity mode to enhanced mode 1
MOD NRDUCELLPAGINGCONFIG: NrDuCellId=0, PagingCapacityMode=ENHANCE_MODE_1;
//Enabling compatibility optimization for the UE release procedure upon UE inactivity
MOD GNODEBPARAM: CompatibilityAlgoSwitch=UE_INACT_REL_PROC_COMPT_OPT_SW-1;
//Enabling the paging origin function
MOD GNODEBALGO: PagingAlgoSwitch=PAGING_ORIGIN_SW-1;
```

Deactivation Command Examples

This section provides an example of how to deactivate the following functions only: OSI payload reduction, RRC message segmentation, RRC connection reestablishment without UE context, paging scheduling, paging interference randomization, and dynamic adjustment of the number of PFs. You may restore the settings of other parameters based on network conditions.

```
//Disabling OSI payload reduction
MOD NRDUCELLDLSCH: NrDuCellId=0, DlSchAlgoSwitch=OSI_PAYLOAD_REDUCE_SW-0;
//Disabling RRC message segmentation
MOD GNODEBPARAM: ProcessSwitch=RRC_MESSAGE_SEGMENT_SW-0;
//Disabling RRC connection reestablishment without UE context
MOD GNODEBPARAM: MobilityOptSwitch=NO_CONTEXT_REEST_SW-0;
//Disabling paging scheduling
MOD NRDUCELLPAGINGCONFIG: NrDuCellId=0, PagingSchAlgoSwitch=PAGING_CR_ADAP_SW-0;
//Disabling paging interference randomization
MOD NRDUCELLPAGINGCONFIG: NrDuCellId=0, PagingSchAlgoSwitch=PAGING_INTRF_RANDOM_SW-0;
//Disabling dynamic adjustment of the number of PFs
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=PAGING_FRAME_NUM_DYN_ADJ_SW-0;
//Disabling compatibility optimization for the UE release procedure upon UE inactivity
MOD GNODEBPARAM: CompatibilityAlgoSwitch=UE_INACT_REL_PROC_COMPT_OPT_SW-0;
//Disabling the paging origin function
MOD GNODEBALGO: PagingAlgoSwitch=PAGING_ORIGIN_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

This basic function takes effect by default and does not require activation verification.

4.4.3 Network Monitoring

This basic function ensures basic service performance. Separate monitoring is not required.

5 Basic Signaling Procedures in NSA Networking

In NSA networking, a UE capable of E-UTRA-NR dual connectivity (EN-DC) can connect to both an eNodeB and a gNodeB, simultaneously using the radio resources provided by these base stations for data transmission. The access process of such a UE includes not only cell search and RA to an LTE cell, but also secondary gNodeB (SgNB) addition and RA to an NR cell. During this process, as control-plane information is carried by the eNodeB in NSA networking, the initial access procedure before SgNB addition takes place in the same way as that in LTE networks. In addition, the process incorporates NR B1 measurement, NG-RAN radio bearer management (including SgNB addition), and RA on the NR side.

Figure 5-1 shows the basic signaling procedures in NSA networking.

Figure 5-1 Basic signaling procedures in NSA networking

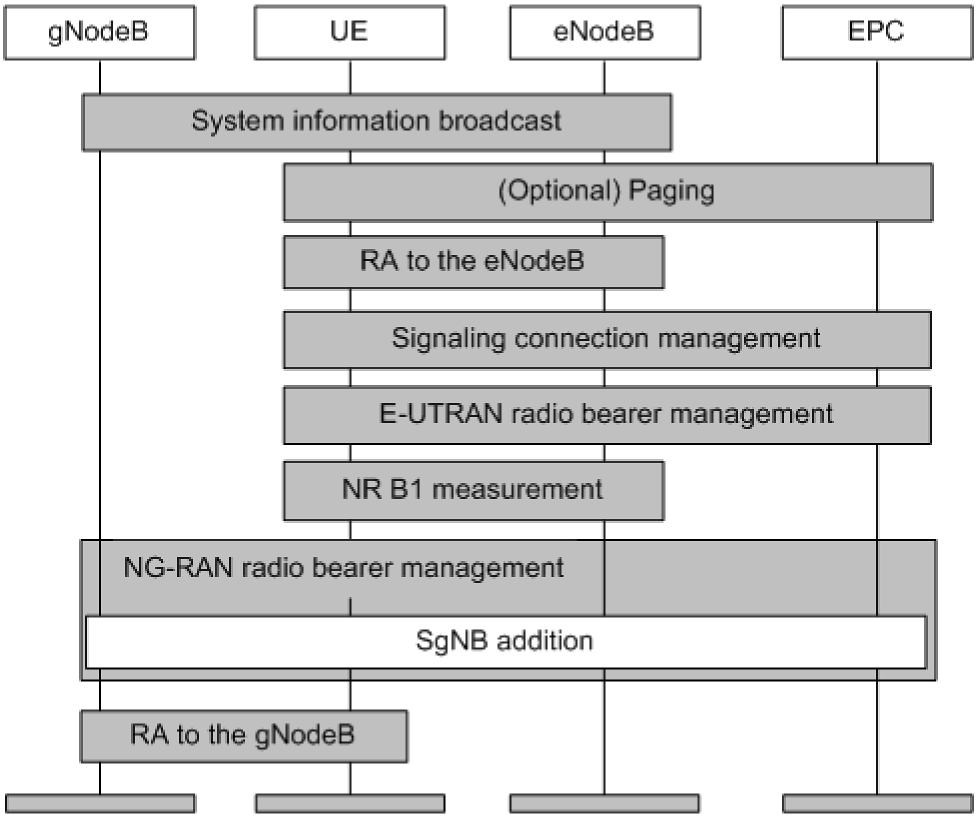


Table 5-1 describes the basic signaling procedures in NSA networking.

Table 5-1 Basic signaling procedures in NSA networking

Procedure	Description	Reference
System information broadcast	System information broadcast is the first step for a UE to obtain the basic network service information. Through this procedure, the UE can obtain the basic AS and NAS information.	**5.1.1 System Information Broadcast**
Paging	When the network needs to set up a connection with a UE, the network initiates a paging procedure to find the UE. This procedure involves the terminating UE, but not the originating UE.	**5.1.2 Paging**
RA to the eNodeB	RA is a mandatory procedure for establishing a radio link between a UE and the network. In NSA networking, the UE sends an access request to the eNodeB, and then the eNodeB responds to the request and allocates a random access channel (RACH).	**5.1.3 RA to the eNodeB**

Procedure	Description	Reference
Signaling connection management	After RA to the eNodeB is complete and before security mode is activated, signaling connections are set up between the UE and the mobility management entity (MME). The connections include an RRC connection and a dedicated S1 connection.	**5.1.4 Signaling Connection Management**
E-UTRAN radio bearer management	E-UTRAN radio bearer management refers to SRB2 and DRB management by the eNodeB after security mode is activated.	**5.1.5 E-UTRAN Radio Bearer Management**
NR B1 measurement	The eNodeB sends a UE the measurement configurations related to event B1 in order to detect the neighboring NR cell with the best signal quality.	**5.1.6 NR B1 Measurement**
NG-RAN radio bearer management	NG-RAN radio bearer management refers to SRB3 and DRB management by the gNodeB.	**5.1.7 NG-RAN Radio Bearer Management**
RA to the gNodeB	A UE sends an access request to the gNodeB.	**5.1.8.1 RA to the gNodeB**

5.1 Principles

5.1.1 System Information Broadcast

In NSA networking, system information broadcast includes system information broadcast on the LTE side and that on the NR side.

- For details about the procedure on the LTE side, see *Idle Mode Management* in *eRAN Feature Documentation*.
- On the NR side, the gNodeB does not broadcast the OSI (because the cell reselection procedure is not involved). It broadcasts MIB to enable UEs to acquire radio frame timing.

5.1.2 Paging

In NSA networking, the UE receives a paging message for originating services on the LTE side, but does not receive any paging message on the NR side. For details about the procedure on the LTE side, see *Idle Mode Management* in *eRAN Feature Documentation*.

5.1.3 RA to the eNodeB

In NSA networking, the initial BWP information is carried in the RRCConnectionReconfiguration message during RA to the eNodeB. Full-bandwidth initial BWP configuration is controlled by the **INIT_BWP_FULL_BW_SW** option (selected by default) of the **NRDUCellAlgoSwitch**.***BwpConfigPolicySwitch*** parameter. Changing the setting of this option will cause the cell to restart, affecting admitted UEs.

- When this option is selected, the gNodeB configures the full-bandwidth initial BWP for all UEs.
- When this option is deselected, the gNodeB configures an initial BWP with the bandwidth specified by CORESET#0 for all UEs, and configures the PRACH frequency-domain position within the initial BWP. In this case, if the frequency-domain position of the initial BWP is not at the edge of cell frequency-domain resources, nor is the PRACH frequency-domain position. This leads to discontinuous PUSCH RBs, affecting the uplink cell throughput. For details about CORESET#0, see section 6.3.2 "Radio resource control information elements" of 3GPP TS 38.331 V15.5.0.

For details about RA to the eNodeB in NSA networking, see *Random Access Control* in *eRAN Feature Documentation*.

5.1.4 Signaling Connection Management

In NSA networking, before SgNB addition, signaling connections include an RRC connection (signaling connection between the UE and eNodeB over the air interface) and a dedicated S1 connection (signaling connection between the eNodeB and MME). For details, see *Connection Management* in *eRAN Feature Documentation*.

NOTE

During SgNB addition, signaling connections also include the X2 connection between the eNodeB and gNodeB. For details, see **5.1.7 NG-RAN Radio Bearer Management**.

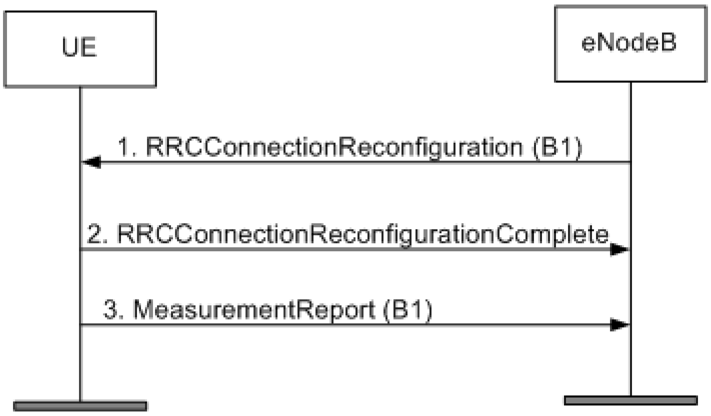
5.1.5 E-UTRAN Radio Bearer Management

Before EN-DC setup, all user plane data is carried by the eNodeB. In this case, radio bearer management involves only E-UTRAN. E-UTRAN radio bearer management refers to SRB2 and DRB management by the eNodeB and involves the setup, modification, and release of SRB2 and DRBs. For details, see *Connection Management* in *eRAN Feature Documentation*.

5.1.6 NR B1 Measurement

In NSA networking, after a UE accesses the network from an eNodeB and selects a neighboring NR cell with the best signal quality based on NR B1 measurement results, the gNodeB serving the NR neighboring cell can work as the SgNB. **Figure 5-2** shows the NR B1 measurement procedure.

Figure 5-2 NR B1 measurement procedure



1. The eNodeB sends an RRCConnectionReconfiguration message including the NR B1 measurement configuration to the UE, requesting the UE to measure neighboring NR cells.
2. The UE sends an RRCConnectionReconfigurationComplete message to notify the eNodeB of the NR B1 measurement completion.
3. The UE sends a measurement report to the eNodeB. The measurement report contains the NR B1 measurement result.

5.1.7 NG-RAN Radio Bearer Management

The gNodeB does not exchange signaling with the UE over the Uu interface. All signaling exchanges between the gNodeB and the UE are forwarded over the X2 interface between the gNodeB and the eNodeB. Therefore, the signaling connections on the NR side involve only X2 connections. After an X2 connection is set up and the gNodeB receives an NR B1 measurement report from the UE, an SgNB addition procedure can be triggered. For details about the SgNB addition signaling procedure, see *EPC-based NSA Basics*, *EPC-based NSA Performance Enhancement*, and *EPC-based NSA Experience Improvement*.

NOTE

When the UE supports SRB3 and the **NSA_DC_SRB3_SWITCH** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter is selected, the gNodeB and UE can exchange signaling using SRB3. For details, see *EPC-based NSA Basics*, *EPC-based NSA Performance Enhancement*, and *EPC-based NSA Experience Improvement*.

During the SgNB addition procedure, the MCG bearer is changed to MCG split bearer or SCG split bearer.

- In Option 3, the eNodeB distributes the user plane data to the gNodeB and itself, and the bearer is called the MCG split bearer.
- In Option 3x, the gNodeB distributes the user plane data to the eNodeB and itself, and the bearer is called the SCG split bearer.

After EN-DC setup, radio bearer management involves NG-RAN. The NG-RAN radio bearer management refers to SRB3 and DRB management by the gNodeB.

- SRB3 management by the gNodeB involves the setup and release of SRB3. For details, see *EPC-based NSA Basics*, *EPC-based NSA Performance Enhancement*, and *EPC-based NSA Experience Improvement*.

- DRB management by the gNodeB involves the setup, modification, and release of DRBs. For details, see the following sections.

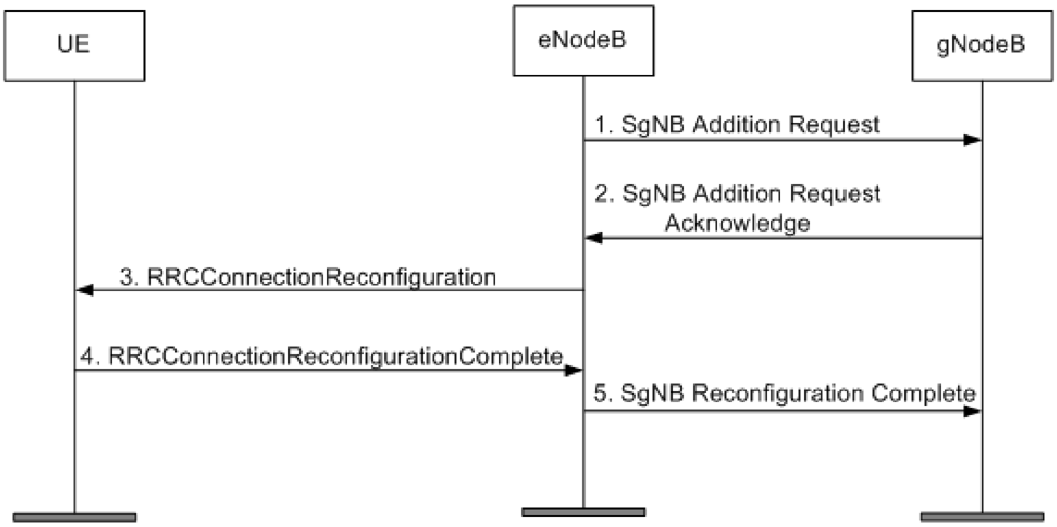
5.1.7.1 DRB Setup

A DRB can be set up after encryption and integrity protection are complete and the UE context is created. DRB setup is triggered when the eNodeB sends an SgNB Addition Request or SgNB Modification Request message. The RRCConncetionReconfiguration message includes a drb-ToAddModList field in the *Radio Resource Config Dedicated* IE. Upon receiving the message, the UE performs the following operations:

- Sets up a PDCP entity and configures related security parameters.
- Sets up and configures an RLC entity.
- Sets up and configures a DTCH.

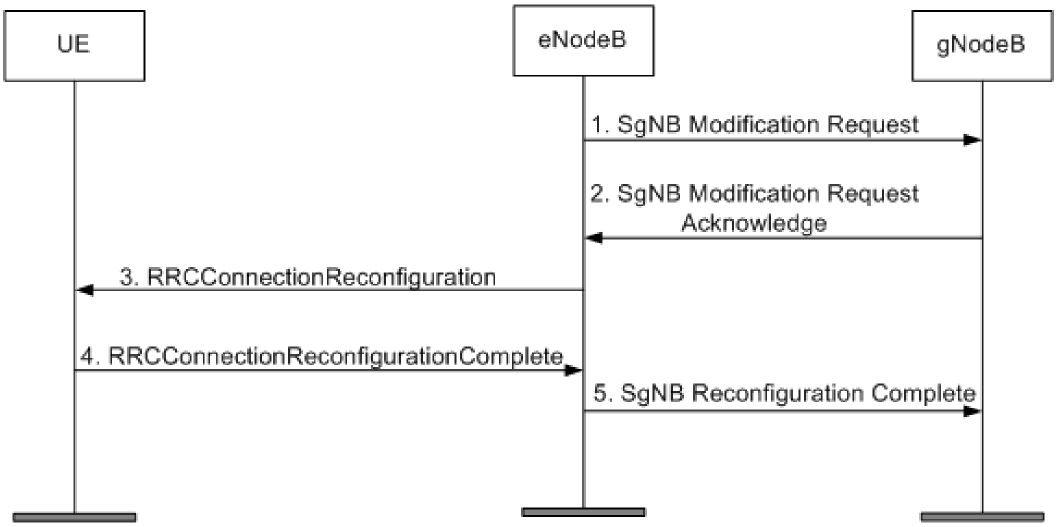
During SgNB addition, the eNodeB sends an SgNB Addition Request message to the gNodeB, instructing the gNodeB to set up a DRB. **Figure 5-3** shows the DRB setup procedure.

Figure 5-3 DRB setup procedure



When a new data split bearer is set up after SgNB addition, the eNodeB sends an SgNB Modification Request message to the gNodeB, instructing the gNodeB to set up a DRB. **Figure 5-4** shows the DRB setup procedure.

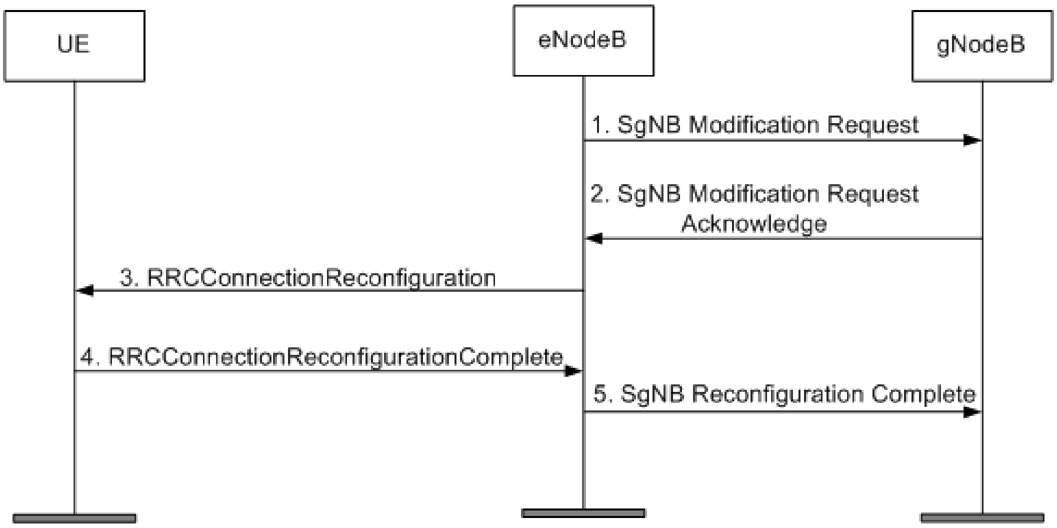
Figure 5-4 DRB setup procedure



5.1.7.2 DRB Modification

DRB modification is triggered when the eNodeB sends an SgNB Modification Request message. **Figure 5-5** shows the DRB modification procedure. According to the instructions in an RRCConnectionReconfiguration message, the UE reconfigures the PDCP entity, RLC entity, and DTCH.

Figure 5-5 DRB modification procedure

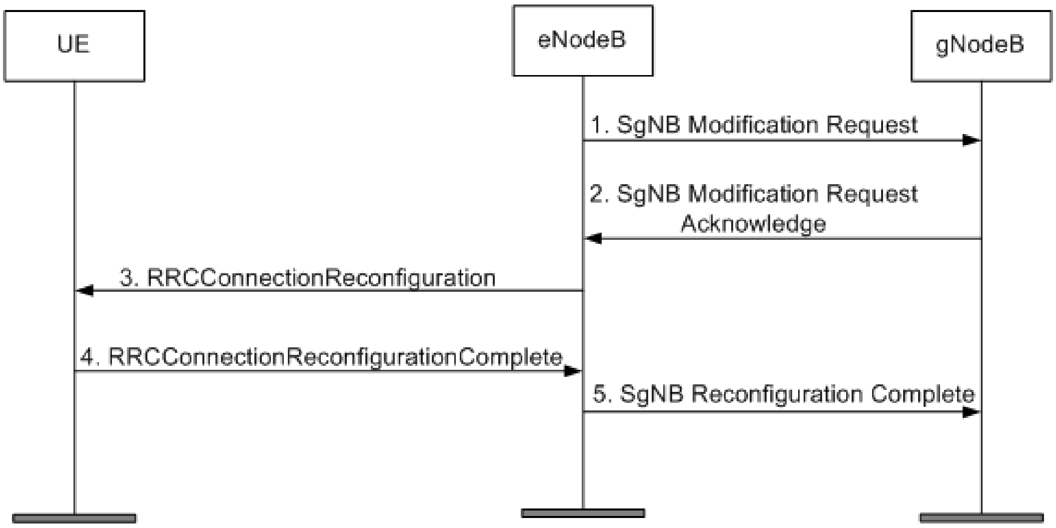


5.1.7.3 DRB Release

During a DRB release, the RRCConnectionReconfiguration message includes a drb-ToReleaseList field in the *Radio Resource Config Dedicated* IE. Based on this message, the UE releases all the resources related to the DRB. A DRB can be released in the following scenarios:

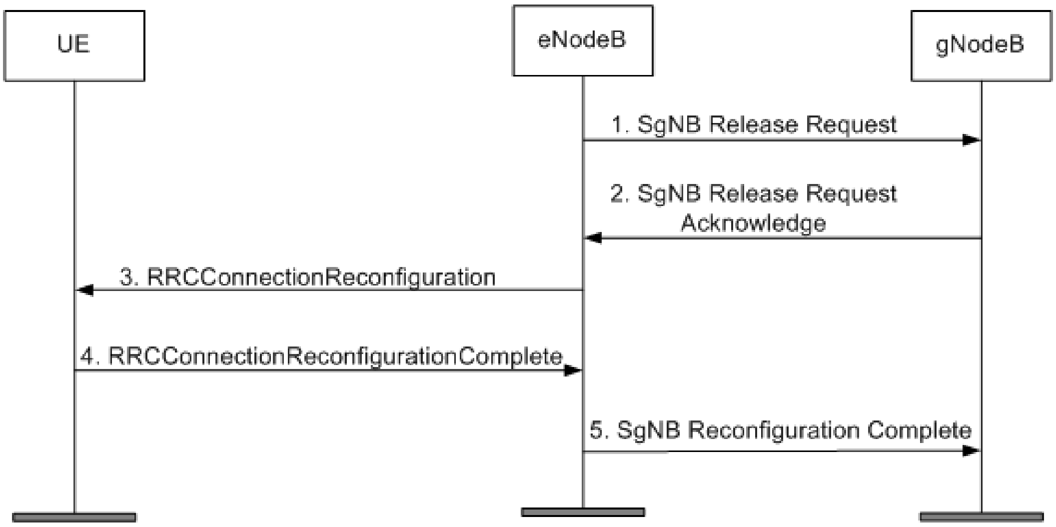
- The eNodeB sends an SgNB Modification Request message. **Figure 5-6** shows the DRB release procedure.

Figure 5-6 DRB release procedure



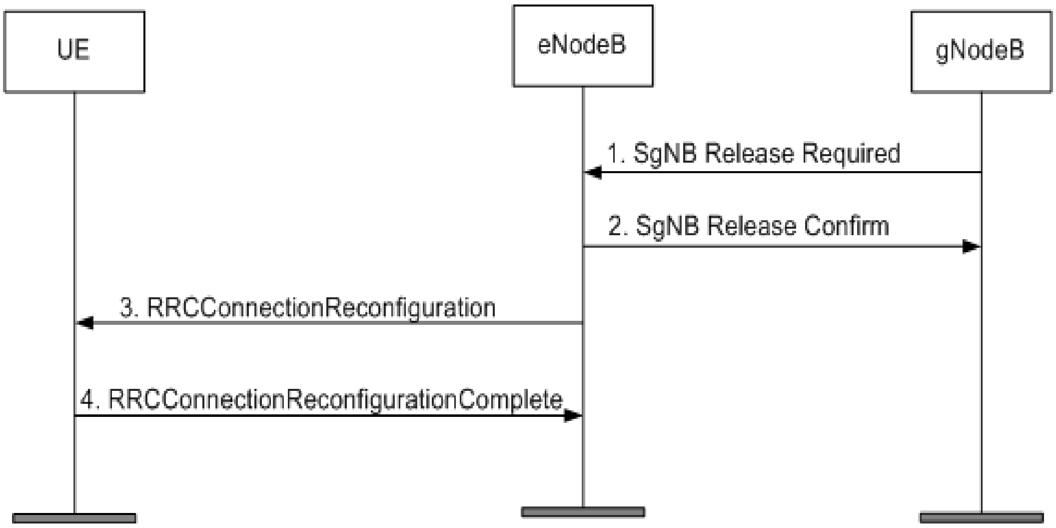
- The eNodeB sends an SgNB Release Request message. **Figure 5-7** shows the DRB release procedure.

Figure 5-7 DRB release procedure



- The gNodeB sends an SgNB Release Required message. **Figure 5-8** shows the DRB release procedure.

Figure 5-8 DRB release procedure



5.1.8 Connection Management on the SgNB Side

5.1.8.1 RA to the gNodeB

For details about RA to gNodeBs in NSA networking, see *Random Access Control*.

5.1.8.2 Uplink Out-of-Synchronization Management

The principles for uplink out-of-synchronization management on the NR side in NSA networking are the same as those in SA networking. For details, see **4.1.4.5 Uplink Out-of-Synchronization Management**.

5.1.8.3 UE Inactivity Management

After detecting an inactive UE, the gNodeB performs inactivity management on the UE. This prevents the inactive UE from occupying system resources for a long time. A UE becomes inactive because of no services or its disconnection from the gNodeB.

After the UE sets up a DRB, the gNodeB considers the UE to be inactive if the UE does not transmit or receive any data (excluding MAC CEs) within the length of the UE inactivity timer (specified by the **NRDUCellQciBearer**.***UeInactivityTimer*** parameter). The UE may set up multiple DRBs, and each DRB is associated with a QCI. If different UE inactivity timer lengths are configured for these QCIs by setting the **NRDUCellQciBearer**.***UeInactivityTimer*** parameter, the maximum of them takes effect.

After the UE becomes inactive, the gNodeB removes the SgNB resources for the UE.

5.2 Network Analysis

5.2.1 Benefits

The functions described in this chapter are basic functions in NSA networking and are enabled by default along with cell activation.

5.2.2 Impacts

The impacts between this function and other functions are described in **5.3.2.3 Function Impacts**.

The function has no impact on the network.

5.3 Requirements

5.3.1 Licenses

None

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Initial CCE aggregation level selection optimization	**AGG_LVL_INIT_SELECT_OPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter	*Channel Management*	The initial CCE aggregation level selection optimization function must be enabled before the full-bandwidth initial BWP configuration function (controlled by the **INIT_BWP_FULL_BW_SW** option of the **NRDUCellAlgoSwitch**.***BwpConfigPolicySwitch*** parameter) and PDCCH rate matching function (controlled by the **PDCCH_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter) are both enabled.

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low latency and high reliability	**HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	The full-bandwidth initial BWP is recommended when uplink interference randomization-based scheduling is enabled for fixed resources of high-reliability services.
High-frequency TDD	None	None	None	None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.4 Operation and Maintenance

The basic NSA networking functions described in this chapter are enabled by default along with cell setup. For details about cell setup in NSA networking, see *EPC-based NSA Basics*, *EPC-based NSA Performance Enhancement*, and *EPC-based NSA Experience Improvement*.

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- 3GPP TS 23.501: "System Architecture for the 5G System"
- 3GPP TS 33.501: "Security architecture and procedures for 5G System"
- 3GPP TR 38.801: "Study on new radio access technology: Radio access architecture and interfaces"
- 3GPP TS 38.300: "NR; NR and NG-RAN Overall Description"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- 3GPP TS 38.321: "NR; Medium Access Control (MAC) protocol specification"
- 3GPP TS 38.304: "NR; User Equipment (UE) procedures in Idle mode and RRC Inactive state"
- 3GPP TS 38.413: "NG-RAN; NG Application Protocol (NGAP)"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.323: "NR; Packet Data Convergence Protocol (PDCP) specification"
- *Cell Management*
- *EPC-based NSA Basics*
- *EPC-based NSA Performance Enhancement*
- *EPC-based NSA Experience Improvement*
- *Channel Management*
- *High Speed Mobility*
- *eXn Self-Management*
- *NG and Xn Self-Management*
- Feature parameter description documents in *eRAN Feature Documentation*:
  - *Idle Mode Management*
  - *S1 and X2 Self-Management*
  - *EPC-based NSA Basics*
  - *EPC-based NSA Performance Enhancement*
  - *EPC-based NSA Experience Improvement*
  - *Random Access Control*
  - *Connection Management*
- *Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*

- *URLLC*
- *Cell Combination*
- *CA Experience Improvement*
- *LTE and NR Power Allocation*
- *CoMP*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *DRX*
- *RedCap*
- *Flow Control*
- *Standards Compliance*
- *Remote Interference Management (Low-Frequency TDD)*
- *UE Power Saving*
- *Energy Conservation and Emission Reduction*